

LLM BASED MENTAL TRAIT EVALUATION FROM ONLINE INTERVIEWS

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Abstract

Personality characterization and mental disorder evaluations, such as for Alexithymia, PTSD, and ADHD, are crucial for accurate diagnostics and effective treatment, guiding treatment plans and optimizing medication. These conditions profoundly impact individuals' well-being and daily functioning, but current diagnostic methods, including clinical interviews and self-report questionnaires, are subjective and often biased. Advanced methods are needed to enhance diagnostic accuracy. This study explores the use of large language models (LLMs) for assessing the severity of mental conditions. We conducted interviews with individuals affected by PTSD, transcribed the conversations, and analyzed the text using LLMs to simulate questionnaire scoring and derive additional insights, comparing these results with traditional self-report questionnaires. Video and audio from interviews were recorded and transcribed, and participants also completed the TAS-20 and IES-R questionnaires. Utilizing prompt engineering, we analyzed the transcriptions with an LLM, and a trained psychologist provided a benchmark assessment for comparison. Our analysis demonstrated that LLMs could consistently simulate questionnaire scoring and provided deeper insights into mental states. Quantitative results showed a high correlation between LLM-based assessments and traditional methods, while qualitatively, LLMs offered nuanced understanding beyond standard questionnaires. This study pioneers the integration of LLMs into mental health evaluations, providing a robust, data-driven approach to enhance traditional diagnostic practices. By leveraging AI, we aim to improve diagnostic accuracy, reduce healthcare costs, and address issues such as AI hallucinations. Our research underscores the transformative potential of AI in mental health assessment, aiming for responsible integration of AI technologies to foster improved patient outcomes and alleviate healthcare professional burnout.

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1. INTRODUCTION

1.1. Mental Disorder Severity Evaluation Background

Personality characterization, and mental disorders evaluation play an important role in psychology. Among the existing mental or personality traits are Alexithymia [1], the impaired ability of recognize or decode emotions, Post Traumatic Stress Disorder (PTSD), debilitating condition initiated by exposure to traumatic events [2], and Attention-Deficit Hyperactivity Disorder (ADHD), inattention and hyperactivity/impulsivity [3]. For all these mental traits and disorders, there is a need to evaluate continuously the severity of the mental condition for more accurate diagnostics, and tailoring the treatment, and optimize medication dose and type.

1.2. Motivation

These conditions can significantly impact an individual's mental health and overall well-being, often manifesting in various symptoms that may overlap or co-occur. Alexithymia, characterized by difficulties in identifying and expressing emotions, can complicate therapeutic interventions and hinder emotional regulation [4]. PTSD, stemming from exposure to traumatic events, can lead to debilitating symptoms such as flashbacks, nightmares, and severe anxiety, profoundly affecting daily functioning and quality of life [5]. Similarly, ADHD, marked by challenges in attention, impulse control, and hyperactivity, can impair academic, occupational, and social success if left untreated [6]. Accurate assessment methods, including comprehensive clinical interviews, standardized rating scales, and psychological testing, are essential for accurately diagnosing these conditions and tailoring treatment plans to individuals' specific needs [7]. Furthermore, timely identification allows for prompt access to appropriate interventions, which may include psychotherapy, medication management, and supportive services [8]. By prioritizing accurate assessment, healthcare providers can optimize resource allocation, reduce unnecessary healthcare costs, and ultimately improve outcomes for individuals affected by these complex mental health conditions [9].

1.3. Problem statement

Currently, there are shortcomings of current diagnosis methods, notably subjective clinical interviews prone to bias and inaccuracy. The complex nature of PTSD for example, with its diverse phenotypic expressions, challenges clinicians in differentiation from other disorders. Additional hurdles include potential feigned diagnoses and concerns about stigmatization. Nevertheless, accurate Alexithymia, PTSD and ADHD assessment is crucial for timely treatment and efficient allocation of healthcare resources [10].

1.4. Existing methods for Mental disorder (Alexithymia, PTSD, ADHD) evaluation disadvantage

Clinical judgment, although commonly employed in the assessment of mental disorders, raises concerns regarding its psychometric validity and interrater reliability

[11]. Present methodologies for evaluating mental disorders predominantly rely on self-report questionnaires. For instance, the Toronto Alexithymia Scale (TAS-20) serves as a prominent tool for assessing alexithymia, with most of the adolescent research on the subject utilizing this instrument [12]. Similarly, the Impact of Event Scale (IES) and its revised version (IES-R) stand out as widely utilized self-report measures for assessing PTSD symptoms [13]. The diagnostic process for ADHD is often intricate, involving a comprehensive collection of behavioral history across various environments [14]. However, reliance solely on self-report instruments and interview administration harbors limitations, as subjects' restricted introspection may skew questionnaire results, and interviews may prove labor-intensive [15], [16]. Clinicians acknowledge biases inherent in self-reports, recognizing limitations stemming from subjects' introspective abilities. This issue is particularly pertinent in the evaluation of Alexithymia, where subjects may lack adequate introspection. Additionally, rating scales introduce biases, as responses may be influenced by individuals' tendencies to exaggerate or provide exciting answers [17]. As technology evolves, recent advancements have extended beyond traditional methodologies towards Machine Learning (ML) and Language Models (LLM) for decoding mental disorders. While conventional approaches like clinical judgment and self-report questionnaires remain fundamental, ML and LLM offer innovative avenues for enhancing diagnostic precision and scalability. The utilization of ML-based approaches for mental disorder screening represents a paradigm shift in diagnostic practices. Leveraging large-scale datasets like the Distress Analysis Interview Corpus of Human and Computer interviews (DAIC_WoZ), researchers have explored the potential of ML algorithms in detecting subtle behavioral patterns indicative of psychological distress [18]. These studies have focused on various modalities, including nonverbal behavior indicators, verbal cues, and sentiment analysis, to uncover hidden markers associated with depression and PTSD [19],[20]. By harnessing the power of ML, researchers can augment traditional diagnostic methods with objective, data-driven insights, improving accuracy and efficiency in mental health assessments.

In parallel, the emergence of Language Models (LLM) has revolutionized our understanding of mental disorders through linguistic analysis. Pretrained on vast corpora of text, LLMs like GPT-3 possess the ability to comprehend and generate human-like language, enabling sophisticated analysis of textual data related to mental health. Recent studies have demonstrated the utility of LLMs in extracting subtle linguistic cues indicative of psychological distress, offering valuable insights into individuals' emotional states and cognitive processes [21]. By analyzing textual data from sources such as clinical interviews, social media posts, and electronic health records, LLMs can uncover latent semantic associations and provide contextually relevant information for diagnostic purposes. However, while ML and LLM offer promising avenues for enhancing mental disorder diagnosis, challenges persist in their implementation. Issues such as data privacy, model interpretability, and generalizability across diverse populations require careful consideration to ensure ethical and effective deployment of these technologies [22], [23]. Moreover, the integration of ML and LLM into clinical practice necessitates

interdisciplinary collaboration between clinicians, data scientists, and ethicists to navigate complex ethical and regulatory frameworks.

1.5. Suggested method (Using video clips from interviews + LLM model)

In our research, we conducted interviews with individuals potentially afflicted by Post-Traumatic Stress Disorder (PTSD) considering the ongoing situation in Israel and the conflict with Gaza. These interviews were held remotely via Zoom meetings, with both video and audio were recorded and transcribed into textual form in Hebrew. Participants were then requested to complete the Toronto Alexithymia Scale (TAS-20) and the Impact of Event Scale-Revised (IES-R) questionnaires to establish a baseline for future evaluations. Employing prompt engineering techniques, the transcribed text underwent analysis by a Language Model (LLM) to discern mental states, simulate questionnaire scoring, and extract additional insights, ensuring consistency in our findings. Additionally, a trained psychologist provided an exemplary assessment to enhance the reliability of our results. Furthermore, we meticulously structured a dataset based on patients' self-reports, utilizing it for fine-tuning the LLMs and aiming to bolster the robustness and accuracy of our findings.

1.6. Research Contribution

Our research endeavors to navigate the unexplored terrain of LLM based psychological assessment, offering a pioneering approach toward leveraging LLMs for the attainment of consistent and reliable results in mental health evaluation. Recognizing the transformative potential of Artificial Intelligence (AI) in medicine, we aim to contribute to the advancement of medical care while mitigating the risks associated with AI-generated content that deviates from reality, known as "AI hallucinations" [24]. By addressing this phenomenon and emphasizing the importance of cautious utilization of AI technologies in healthcare, our study seeks to pave the way for responsible integration of AI into psychological assessment practices, thereby fostering improved patient outcomes and alleviating healthcare professional burnout.

2. EXPERIMENTAL SETUP

2.1. Experiment Setup Description

The recordings from these interviews encompass both audio and video files. Initially, the analysis focused exclusively on the auditory component, necessitating the utilization of speech-to-text transcription tools. Figure 2 describes the experiment setup



Figure 1 - Experiment setup

2.2. Clinical labeling

In the mental evaluation we use we were predicting – PTSD and Alexithymia - there are 3 symptom types, and typically each question in the questionnaires is measuring one of the symptoms. The sum of measurements of each symptom is referred to as “**Sub-scores**”. The following mental conditions are:

- **Alexithymia**

The questionnaire was from TAS-20 [25].

The questionnaires are given in [Appendix 8.A](#).

- **PTSD**

The questionnaire was from IES-R [26].

The questionnaires are given in Appendix 8.B.

- **ADHD**

The questionnaire was from ASRS [27].

The questionnaires are given in Appendix 8.C.

3. LLM DESIGN

This chapter explains the considerations that influenced our employment of LLMs, which components were necessary or considered for the prompt design, and what was the extracted data.

3.1. Requirements

The method we employ has to fulfill multiple requirements:

Table 1 - LLM Utilization Requirements

Requirement	Implementation
Low Computational Resources	We will pre-trained state-of-the-art LLMs provided by commercial vendors. The output will be tuned by using Prompt Engineering techniques.
Modularity	In order to efficiently test various permutations of the prompt, the prompt will be constructed in a modular fashion.
Accuracy	The assessment will be compared to the results of the questionnaires.
Consistency	Each prompt will be tested multiple times to verify the average performance.
Reliability	Compare between different prompts, and with the feedback related to certainty from the LLM model

The main takeaway is that the optimization will be done by improving the prompt:

3.2. Prompt Design Considerations

“Does the patient have PTSD?” is a straightforward, yet rudimentary approach to our effort. Such a simplistic inquiry lacks explainability, rendering the outcomes ambiguous, and makes it difficult to assess the accuracy of its assessment. To ensure reliable and interpretable results, the LLM must be instructed to provide a structured response, including a scale, reasoning, confidence and format. To align with established clinical practices, the LLM should evaluate the presence of individual symptoms, as defined by standardized questionnaires. LLMs tend to employ common scoring scales (e.g. 0 to 10, 1 to 10, 0 to 100) which are different from those used in clinical questionnaires. A linear transformation is necessary to adjust the LLM's output to match the scale of the corresponding questionnaire. See Eq. (4.5.1). Instructing the LLM to “act” or “reason” as a mental health expert can enhance the accuracy and relevance of its responses. This encourages the LLM to consider the nuances of clinical language and decision-making processes. Beyond predicting questionnaire scores, the LLM should be prompted to identify specific indicators within the interview that support its assessment. This provides valuable insights into the patient's mental state and aids clinicians in interpreting the results, and assist in the assessment process for the LLM. LLMs can sometimes generate fabricated information, especially

when asked to evaluate conditions based on questionnaires they haven't been explicitly trained on. This is known as "hallucination." Despite being aware of the existence of the questionnaires, they don't possess the knowledge about the content of the questions. Therefore, the full questionnaire should be incorporated into the prompt. The LLM is then tasked with predicting the answer to each question individually and providing a rationale for its prediction. Questionnaires designed for human use may not be optimal for LLM processing. Humans are prompted to mark their answers on a table with a Likert-type scale. Reformatting the questionnaire into a more concise and structured layout can improve the LLM's ability to parse and interpret the questions. Thus, the provided version details the Likert-type scale in the beginning of the questionnaire and then the series of questions. LLMs may resort to neutral or non-committal answers when uncertain or lacking enough context or information in the interview transcript.

For example:

- "3. Neither Agree nor Disagree" as in "I don't know" for TAS-20 prediction
- "0. Not at All" as in "Not mentioned" for IES-R prediction.

This can hinder the diagnostic value of the assessment. The prediction can inflate or flatten the total score. Requiring the LLM to provide a confidence level for each answer allows for a nuanced interpretation of its predictions. Additionally, prompting the LLM to include remarks or explanations for its scoring enhances transparency and facilitates verification if needed. Instructing the LLM to break down its reasoning process step-by-step can lead to more robust and evidence-based assessments. This encourages the LLM to focus on specific details within the interview rather than relying solely on generalized "Intuitive" knowledge. Providing the LLM with examples of interviews and corresponding expert evaluations can guide its understanding of the task and expected response and improve the consistency and quality of its output. This is also necessary for processing the responses later. Using clear tags (e.g. <Example Interview>) to delineate different sections of the prompt helps the LLM maintain structural awareness and avoid confusion between instructions and content. Commercial LLMs often have built-in safeguards to prevent them from offering medical advice. Including a disclaimer that explicitly states the LLM's output is intended as a decision support tool, not a clinical diagnosis, can help circumvent these restrictions.

3.2.1. Additional Considerations Per Mental Conditions

Each mental condition holds its own nuances and challenges. Adjusting to the implications of each condition is necessary to evaluate the patients appropriately.

PTSD

When evaluating PTSD according to the IES-R questionnaire, a key sentence is essential for the assessment: *"How distressing each difficulty has been for you during the past seven days with respect to the event"*. When attempting to predict the scores for the items in the questionnaire, the LLM must make the distinction between what the patient is projecting during the interview, and how he felt in days following the traumatic event.

3.3. Prompt Design

3.3.1. Definitions:

Throughout the project we coined multiple terms

Speaker Sampling:

The transcription service we used had a built-in speaker recognition feature and has split the transcript to **spk_0** and **spk_1**. As we discussed before, the way we present the interview could possibly have an impact on the output. Thus, we were benchmarking alternative labels for the speakers, but kept using the term “Speaker 1” to refer to the patient.

- **Speaker 0** – Therapist, Interviewer
- **Speaker 1** – Patient, Interviewee, Subject

Additionally, we tested whether excluding the therapist’s questions from the transcript impacted performance.

- **Speaker 0, 1** - Interview transcript including both speakers
- **Speaker 1 only** - Interview transcript without the therapist’s questions.

Prompts types:

We decided to benchmark different approaches and levels of sophistication of the prompts. The prompt types were labeled based on their reliance on questionnaires:

- **Direct** - Attempting to evaluate the final condition score directly without references.
- **Sub-score** - Attempting to evaluate each sub-score of the condition separately, without references.
- **Items** - Attempting to predict the condition score according to the commonly used self-administered questionnaire for that condition. In order to "stick to the source", we predict each question (item) in the questionnaire.

3.3.2. Prompt Components

Based on the prompt design considerations in 3.1. we devised a function that would generate a modular prompt ("Prompt Builder") to incorporate multiple improvements to the naïve approach. We call those improvements **Prompt Components**, and their variant was passed as a parameter (function argument) to the prompt builder. Each prompt may include the following components:

Table 2 - Prompt Components

Prompt Component	Details
Instructions	General instructions, instructing the LLM to assume the role of a mental health expert.

Prompt Component	Details
Context	Background information defining key terms, assessment goals, and diagnostic criteria.
Chain of thoughts segments	Step-by-step instructions guiding the LLM through a structured reasoning process.
Questionnaire	The full text of the relevant standardized questionnaire (TAS-20, IES-R, or ASRS), incorporated directly into the prompt.
Example Interview and Example Evaluation	A sample interview transcript and a demonstration of the expected output format.
Interview	The transcript of the interview for assessment.
Disclaimer	A statement clarifying the usage of the assessment “This assessment is intended as a decision support tool and is not a substitute for a clinical diagnosis by a qualified mental health professional.”

3.4. Prompt Output Format

Each iteration, we choose a combination of parameters for the prompt builder, and send the prompt with the interview to the LLM via API calls. The received response is saved in its raw text form, and analyzed in batches later. From the text output we extract the relevant information and store it as data tables.

We kept track of the prompt structure that was used in each iteration, as the different prompt types produce different information, and the predicted scores from each iteration.

3.4.1. Permutation Parameters Structure

For the prompt structure we kept track of the parameters we used to generate the prompt:

Parameter	Details
Prompt Type	Direct, Sub-score, Items
Condition	The mental condition or disorder that is evaluated. (Alexithymia, PTSD, ADHD)
Questionnaire	TAS-20, IES-R, ASRS
Q Format	The format of the questionnaire. (Compact, Detailed)
Instructions	The variant of the instruction

Parameter	Details
Context	The variant of the context.
Chain of Thoughts	The variant of the detailed instructions.
Example Interview	The variant of the provided transcript example. Includes the speaker 1 only variation as well
Example Evaluation	The variant of the evaluation example. Conditions the LLM to output the answer in a certain format
Mental State	Part of the example evaluation component. Whether the answer should include a list of Mental State Indicators
Prediction Output Format	CSV, Natural Language
Disclaimer	Whether to include a disclaimer at the end of the prompt
Batch	A grouping index for the predefined permutations
Unique Index	A unique enumerator for the attempted iteration

3.4.2. Direct Prompt Results

For the results of the direct prompt type we kept track of:

Parameter	Details
File Number	Subject Number
Level	Condition Score
Unique Index	
Confidence Level	
Batch	

3.4.3. Sub-Score Results

For the results of the sub-score prompt type we kept track of:

Parameter	Details
File Number	
Subscore_1	The score of the first symptom of the condition. The score is normalized from 1-10 to the corresponding scale in the original questionnaire
Subscore_1 Confidence	The confidence level for the first symptom.
Subscore_2	

Parameter	Details
Subscore_2 Confidence	
Subscore_3	
Subscore_3 Confidence	
Total Score	The sum of all the symptoms scores, also referred to as the condition score. Transitively normalized to the questionnaire scale
Unique Index	
Batch	

3.4.4. Items Results

For the results of the items prompt type we kept track of:

Parameter	Details
File Number	
Item	The number of the question in the questionnaire
Score	The predicted score to the question
Confidence	The confidence level to the question. On a scale 0-1
Remark	Relevant remarks and reasoning for scoring the prediction. Plain text
Unique Index	
Batch	

4. DATA ANALYSIS

The data analysis is composed of the following stages: Transcription, LLM classification, post processing, and diagnostics. Figure 2 shows the main analysis scheme.

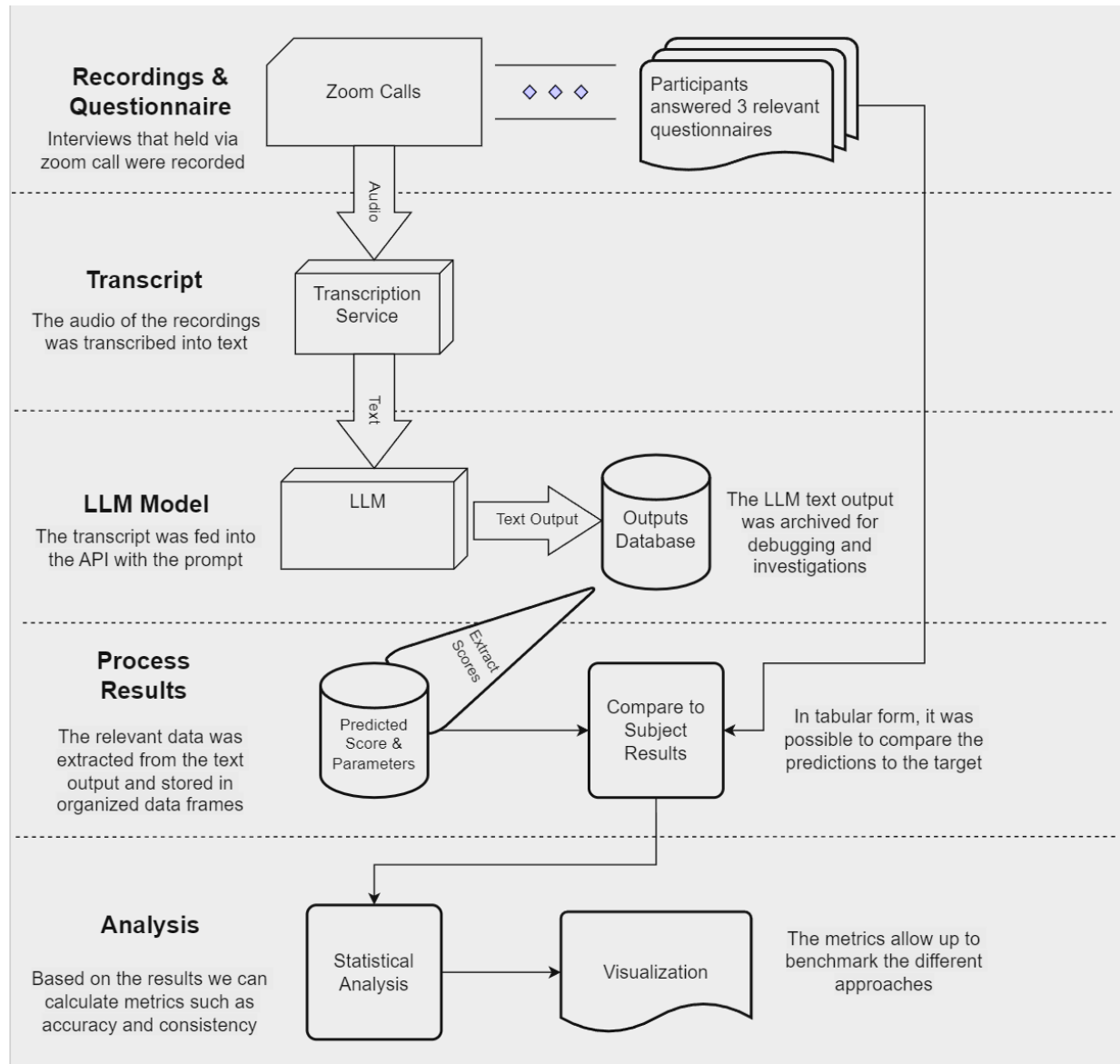


Figure 2 - Model Workflow

4.1. Data Processing

4.1.1. Transcription

A manually transcribed text file served as the benchmark against which various transcription tools, including Google Transcribe, Amazon Transcribe, Vibe (a pretrained Hebrew transcription model), and VeedAI, were evaluated. However, due to discrepancies in output formats among these tools and the ground truth, assessing transcription accuracy proved challenging. As a result, emphasis shifted towards evaluating confidence level percentages provided by advanced tools, such as Google

and Amazon, based on the number of accurately transcribed words. Among these, Amazon demonstrated superior performance, leading to its selection for further analyses. Nonetheless, challenges arose in interpreting Amazon's output, particularly its limitation of visual text on the site interface to 5000 characters, necessitating the download of a JSON file. To address this issue, the "tscribe" Python library was identified, enabling the comprehension of Amazon Transcribe's specific output format and facilitating the creation of a comprehensible document file with confidence level indicators and appropriately segmented speaker transcriptions.

The interviews were held via zoom meetings, both participants had cameras on, and the meeting was recorded and saved to the cloud.

We used the audio of the interview and transcribed it into text using amazon transcribe service, which also made the separation between the speakers.

4.2. LLM Utilization

We utilized the Gemini API, which is designed to facilitate natural language processing tasks, through Python. By sending prompts to the API, we received text responses that were further processed using techniques such as regular expressions to automate the analysis.

After receiving the responses from the API, we processed the data and saved it into text files and corresponding Excel files. These files were then analyzed to evaluate the effectiveness of various prompt techniques. This analysis allowed us to identify the most efficient approach for generating accurate results.

4.3. Hypothesis

Our hypothesis holds that it is possible to achieve a reasonably accurate estimate of a patient's mental state using an LLM-based approach. The accuracy of this estimate is closely tied to the quality of the information provided during the interview. For instance, if the patient offers detailed insights into their recent emotions and feelings, the LLM will have a richer context to analyze, leading to more precise recognition of any underlying mental conditions. Conversely, if the interview lacks depth or emotional disclosure, the model's ability to assess the mental state may be compromised. Thus, the effectiveness of this approach is contingent on both the quality of the interaction and the model's ability to interpret nuanced emotional cues. This underscores the importance of fostering open and thorough communication during the interview process to maximize the reliability of the assessment.

4.4. Reference system

To evaluate the model, we use the questionnaire self-report as reference. For this we need to do sub-score adjustment, then we define the evaluation score

4.5. Sub-score adjustment

LLMs tend to employ common scoring scales (e.g. 0-10, 1-10, 0-100) which are different from those used in clinical questionnaires. A linear transformation is necessary to adjust the LLM's output to match the scale of the corresponding questionnaire.

Let $T: A \rightarrow B$ be a linear transform between scale A to scale B .

$$\begin{cases} T: [A_{min}, A_{max}] \rightarrow [B_{min}, B_{max}] \\ T(x) = B_{min} + (x - A_{min}) \cdot \frac{(B_{max} - B_{min})}{(A_{max} - A_{min})} \end{cases} \quad \text{Eq. (4.5.1):}$$

Where A_{min}, A_{max} , are the lower and upper bounds of the original scale, likewise B_{min}, B_{max} are the lower and upper bounds of the target scale. x is the score we want to transform and $T(x)$ is the corrected score.

For example, to transform the predicted *score* for Alexithymia to the correct scale, we apply the transform:

$$\begin{cases} T_{alexithymia}: [0,10] \rightarrow [20,100] \\ T_{alexithymia}(score) = 20 + \frac{(score-1) \cdot (100-20)}{(10-1)} \end{cases} \quad \text{Eq. (4.5.2):}$$

Likewise:

$$\begin{cases} T_{PTSD}: [0,10] \rightarrow [0,88] \\ T_{PTSD}(score) = \frac{(score-1) \cdot 88}{(10-1)} \end{cases} \quad \text{Eq. (4.5.3):}$$

$$\begin{cases} T_{ADHD}: [0,10] \rightarrow [19,90] \\ T_{ADHD}(score) = 18 + \frac{(score-1) \cdot (90-18)}{(10-1)} \end{cases} \quad \text{Eq. (4.5.4):}$$

Similarly, when we predict the individual sub-scores for each condition, we apply a transform. Each sub-score has its scale, and thus:

Alexithymia Sub-scores

DIF (Difficulty Identifying Feelings):

$$\begin{cases} T_{DIF}: [0,10] \rightarrow [5,25] \\ T_{DIF}(score) = 5 + \frac{(score-1) \cdot (25-5)}{(10-1)} \end{cases} \quad \text{Eq. (4.5.5):}$$

DDF (Difficulty Describing Feelings):

$$\begin{cases} T_{DDF}: [0,10] \rightarrow [7,35] \\ T_{DDF}(score) = 7 + \frac{(score-1) \cdot (35-7)}{(10-1)} \end{cases} \quad \text{Eq. (4.5.6):}$$

EOT (Externally Oriented Thinking):

$$\begin{cases} T_{EOT}: [0,10] \rightarrow [8,40] \\ T_{EOT}(score) = 8 + \frac{(score-1) \cdot (40-8)}{(10-1)} \end{cases} \quad \text{Eq. (4.5.7):}$$

PTSD Sub-Scores

Intrusion:

$$\begin{cases} T_{Intrusion}: [0,10] \rightarrow [0,32] \\ T_{Intrusion}(score) = \frac{(score-1) \cdot 32}{(10-1)} \end{cases} \quad \text{Eq. (4.5.8):}$$

Avoidance:

$$\begin{cases} T_{Avoidance}: [0,10] \rightarrow [0,32] \\ T_{Avoidance}(score) = \frac{(score-1) \cdot 32}{(10-1)} \end{cases} \quad \text{Eq. (4.5.9):}$$

Hyperarousal:

$$\begin{cases} T_{Hyperarousal}: [0,10] \rightarrow [0,24] \\ T_{Hyperarousal}(score) = \frac{(score-1) \cdot (40-8)}{(10-1)} \end{cases} \quad \text{Eq. (4.5.10):}$$

4.6. Performance metrics (measures)

4.6.1. Regression Accuracy

The accuracy of a prediction y' is measured against the real score Y , and is calculated by subtracting the error ϵ from 1.

$$\epsilon(y) = \left| \frac{Y-y'}{B_{max}-B_{min}} \right| \quad \text{Eq. (4.6.1.1):}$$

$$Accuracy = 1 - \epsilon$$

4.6.2. Consistency (Reliability)

The consistency of the answer can be estimated by different means: How much the question is consistent in each iteration per item, or per sub-score.

We calculated the Consistency errors as the coefficient of variation.

The coefficient of variation (CV) is defined as the ratio of the standard deviation σ to the mean μ . calculated as:

$$Coefficient\ of\ Variation = \frac{\sigma}{\mu} \quad \text{where:} \quad \text{Eq. (4.6.2.1):}$$

- Mean (μ): $\mu = \frac{(\sum x)}{n}$
- Standard Deviation (σ): $\sigma = \sqrt{\left[\frac{(\sum (x - \mu)^2)}{n-1} \right]}$
- $\sum x$ is the sum of all values in the dataset

- x represents each value in the dataset
- μ is the mean of the dataset
- n is the number of values in the dataset

4.6.3. Confidence

Confidence of the answer reflects how much the LLM sure in his answer in scale 0-1.

5. STATISTICS DISTRIBUTION

5.1. Statistics Distribution - Alexithymia

5.1.1. Per item (question)

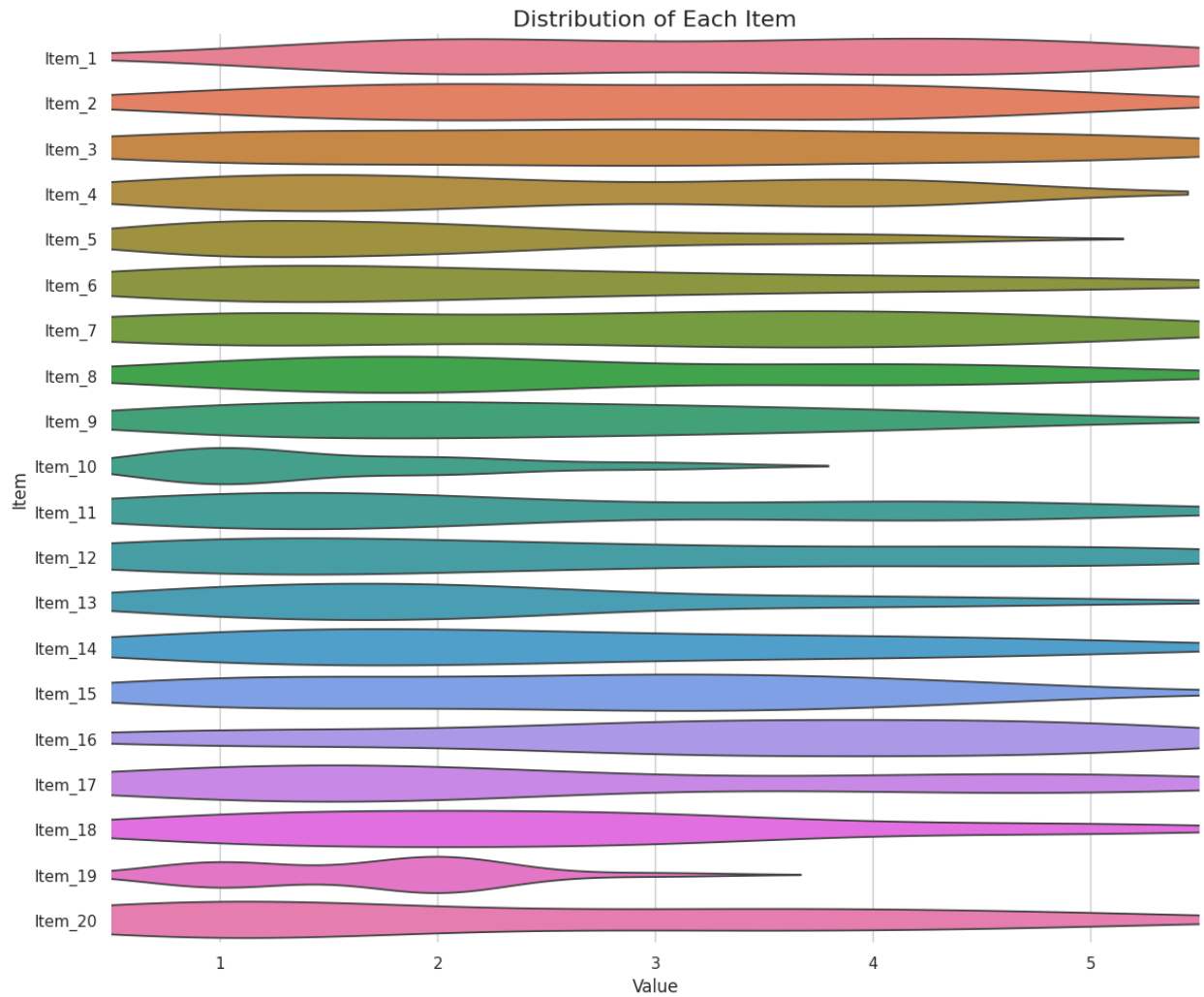


Figure 3 - Alexithymia items

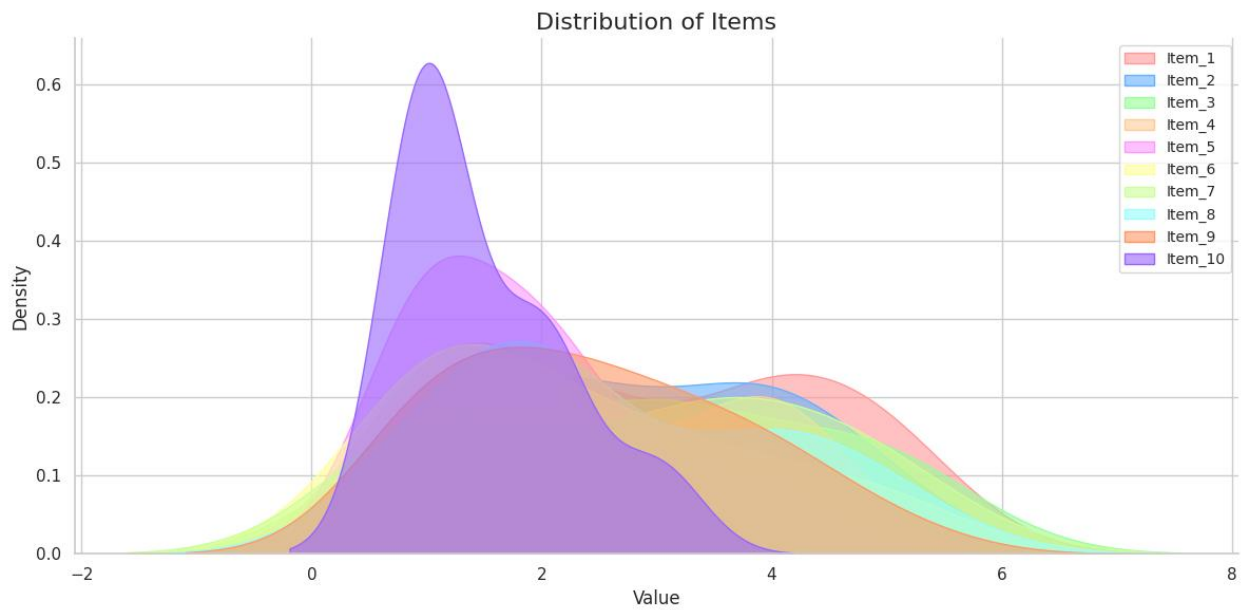


Figure 4 - Alexithymia items 1-10

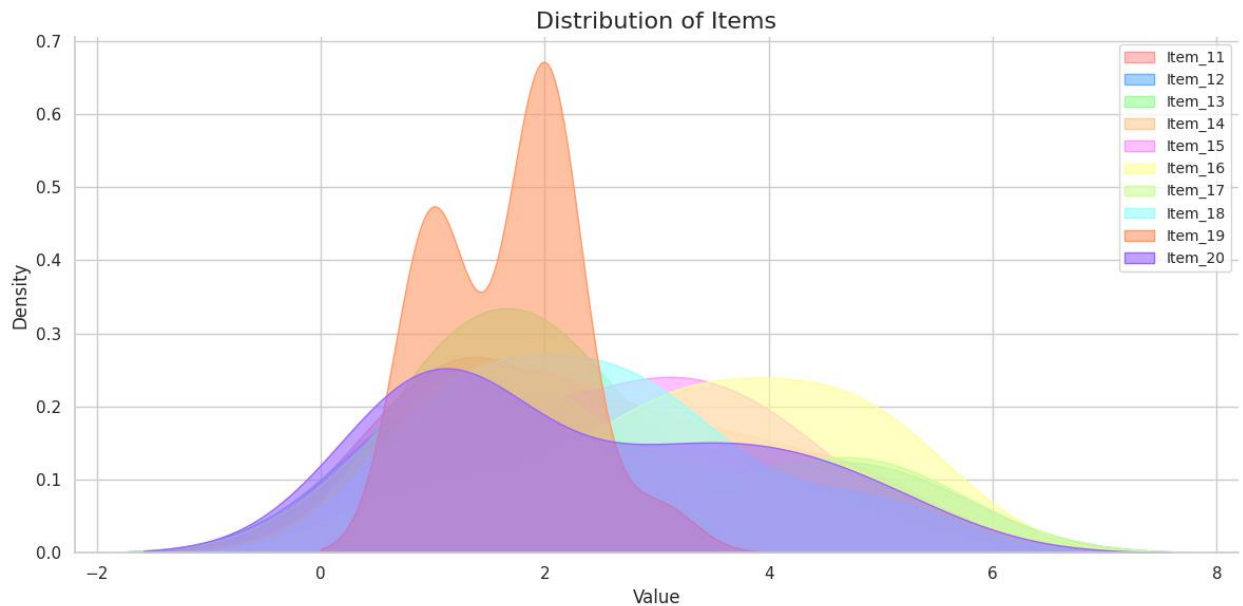


Figure 5 - Alexithymia items 11-20

The distributions of values across the 20 items show considerable variation as we can see in figures 3,4,5. While some items (like 7, 8, and 14) have relatively uniform distributions across the value range, others (such as 10 and 19) display more concentrated peaks, and items like 1, 2, and 20 seem to have higher frequencies at the extremes of the scale.

5.1.2. Per subscale

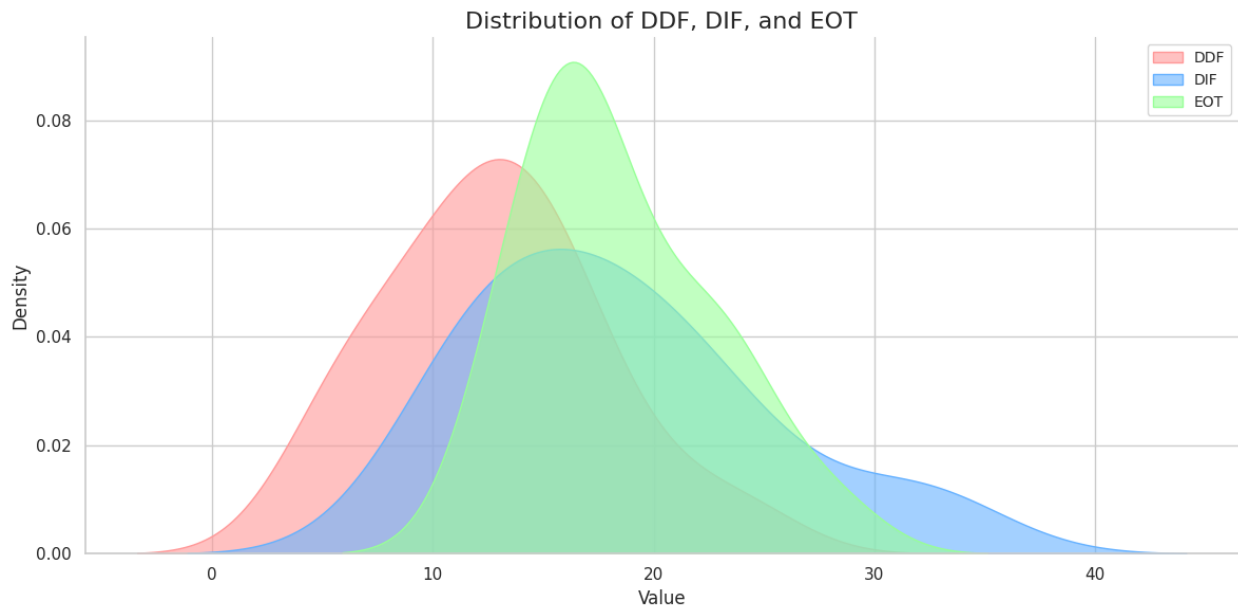


Figure 6 - Alexithymia sub scores

Figure 6 reveals several key insights about the distributions of DDF, DIF, and EOT values. The DDF distribution (pink) is right-skewed, with a peak around 13, indicating that most DDF values are clustered in the lower range, with fewer observations in the higher range. The DIF distribution (green) displays a more symmetrical, bell-shaped curve, peaking around 17, suggesting a more even distribution of DIF values across the range. In contrast, the EOT distribution (blue) is the most widespread, with a flatter, more dispersed curve, indicating a broader range of EOT values without a clear peak or concentration.

5.2. Statistics Distribution - PTSD

5.2.1. Per item (question)

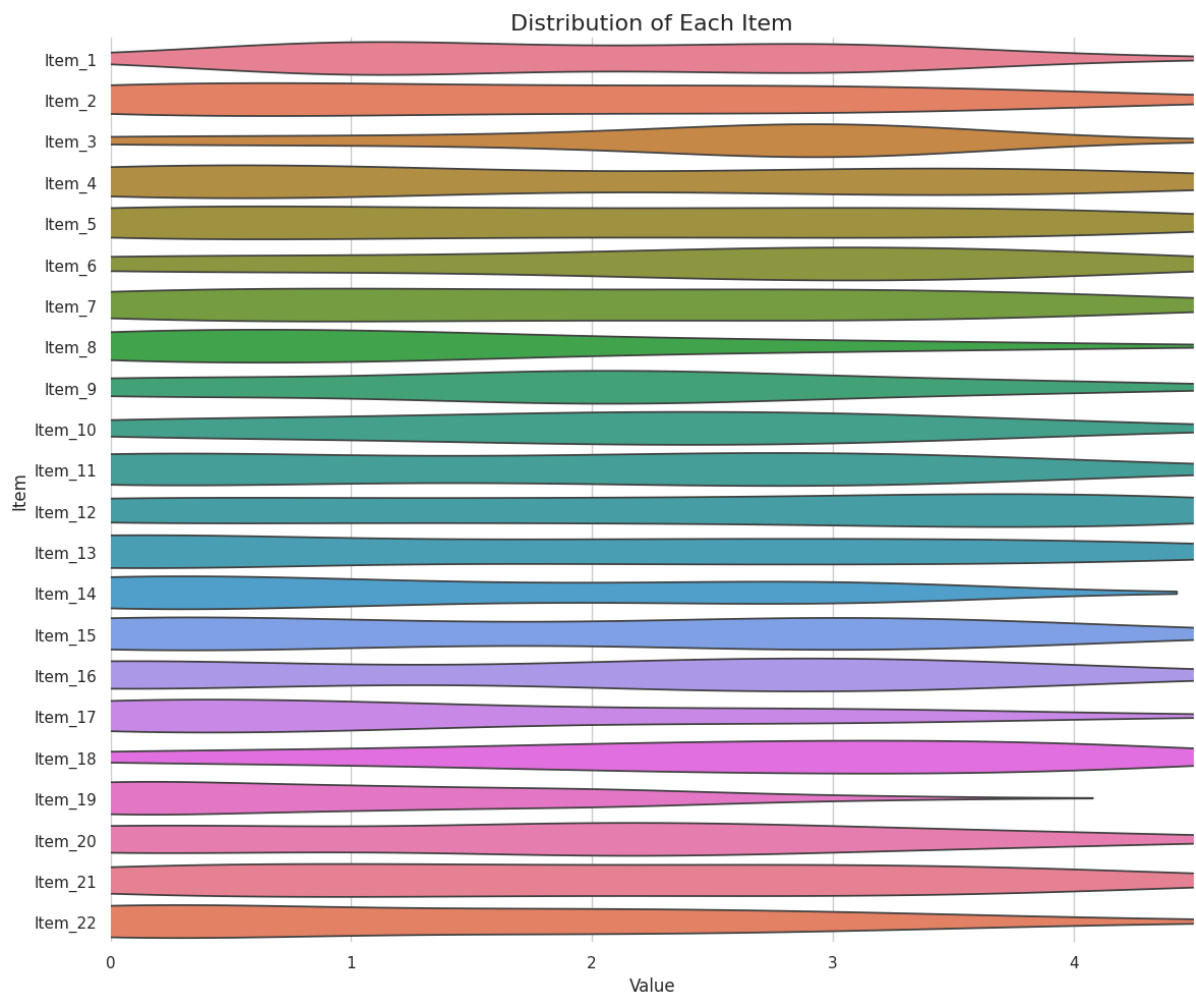


Figure 7 - PTSD items

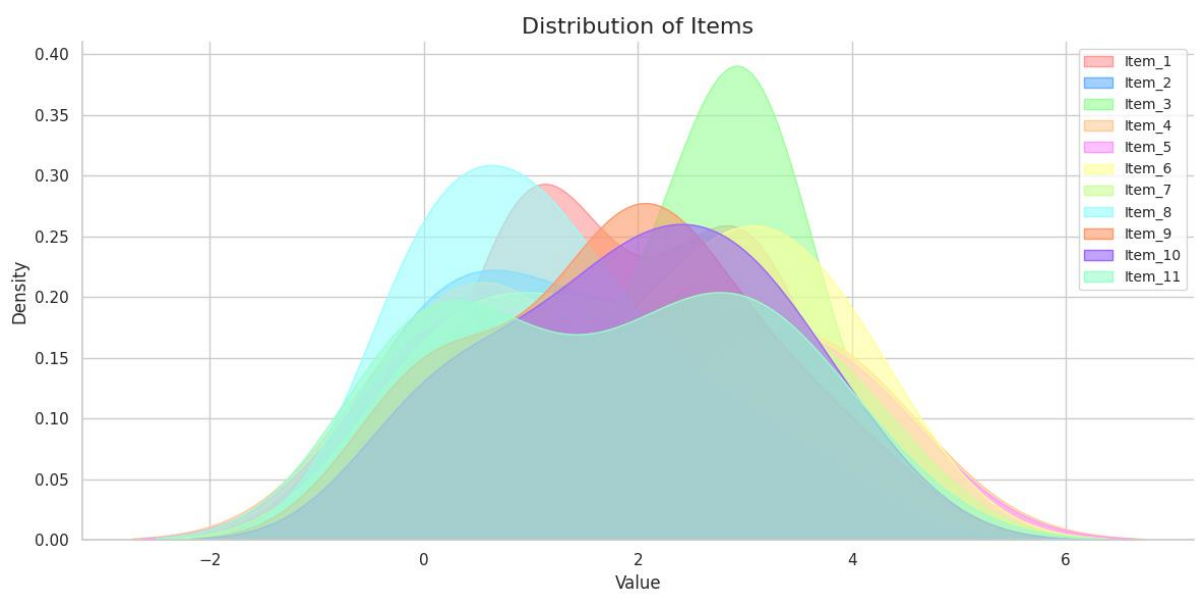


Figure 8 - PTSD items 1-11

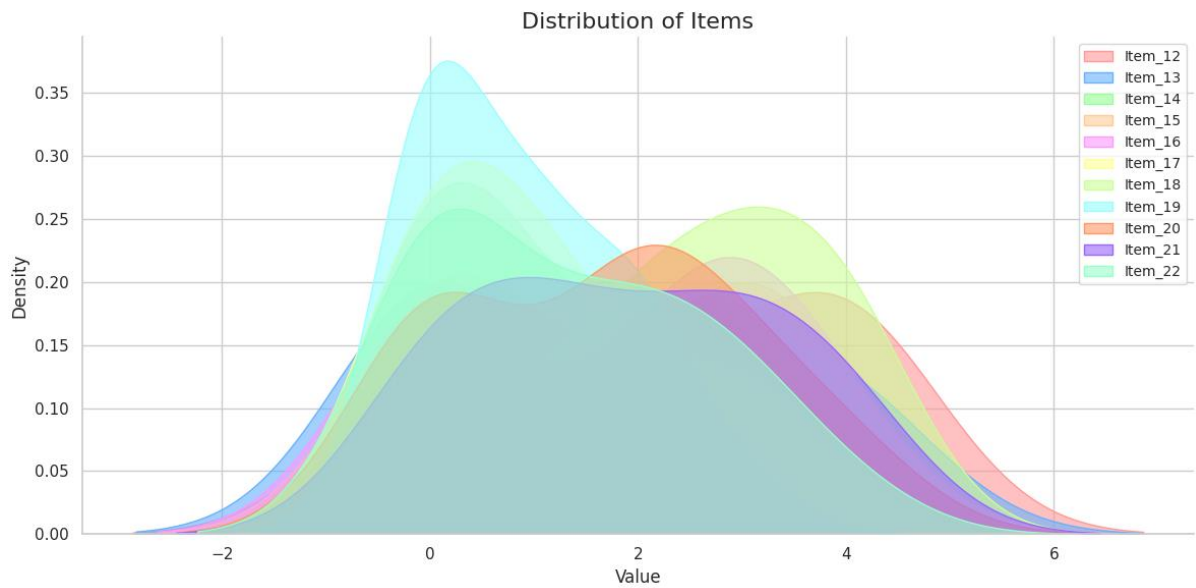


Figure 9 - PTSD items 12 - 22

As we can see in figures 6,7,8 the distributions of values across the 22 items show minimal variation, with most items displaying remarkably uniform distributions across the value range from 0 to 4. While a few items (such as 19) show slight deviations with minor peaks near the center, the vast majority, maintain consistent frequencies across the entire scale, without notable concentration at the extremes or middle values.

5.2.2. Per subscale

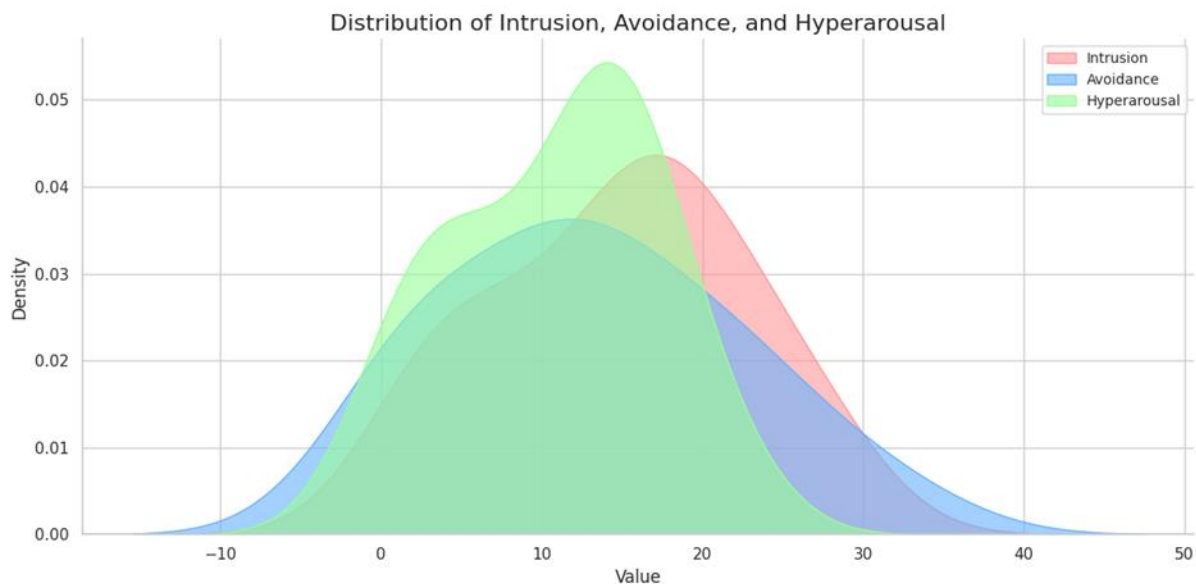


Figure 10 - PTSD sub scores

Figure 10 illustrates the distributions of Intrusion, Avoidance, and Hyperarousal values across a range of data. The Intrusion distribution (red) has a narrower spread and is right-skewed, peaking around a value of 20, indicating that most Intrusion scores are

concentrated in the lower to mid-range with a gradual decline in higher values. The Avoidance distribution (blue) is more dispersed, with a peak near 10 and a noticeable spread across a broader range, suggesting a more even distribution of Avoidance scores. The Hyperarousal distribution (green) is the most prominent, displaying a symmetrical bell-shaped curve that peaks around a value of 15, reflecting that most Hyperarousal scores are concentrated around this central value, with a gradual tapering on either side. Overall, this graph highlights the differences in how these three variables are distributed within the dataset, with Hyperarousal showing the most symmetry, Intrusion being skewed towards lower values, and Avoidance having a wider distribution.

5.3. Statistics Distribution - ADHD

5.3.1. Per item (question)

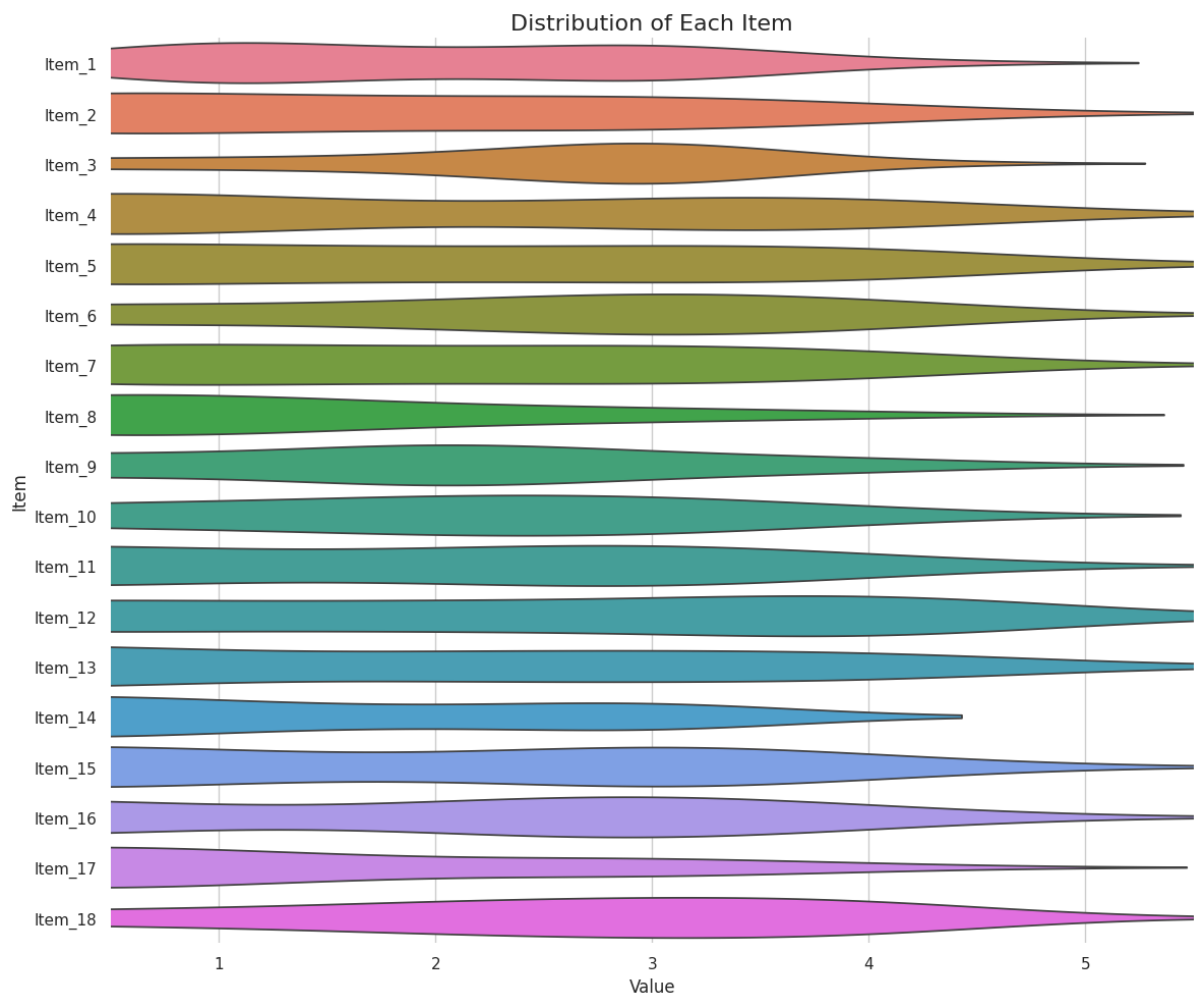


Figure 11 - ADHD items

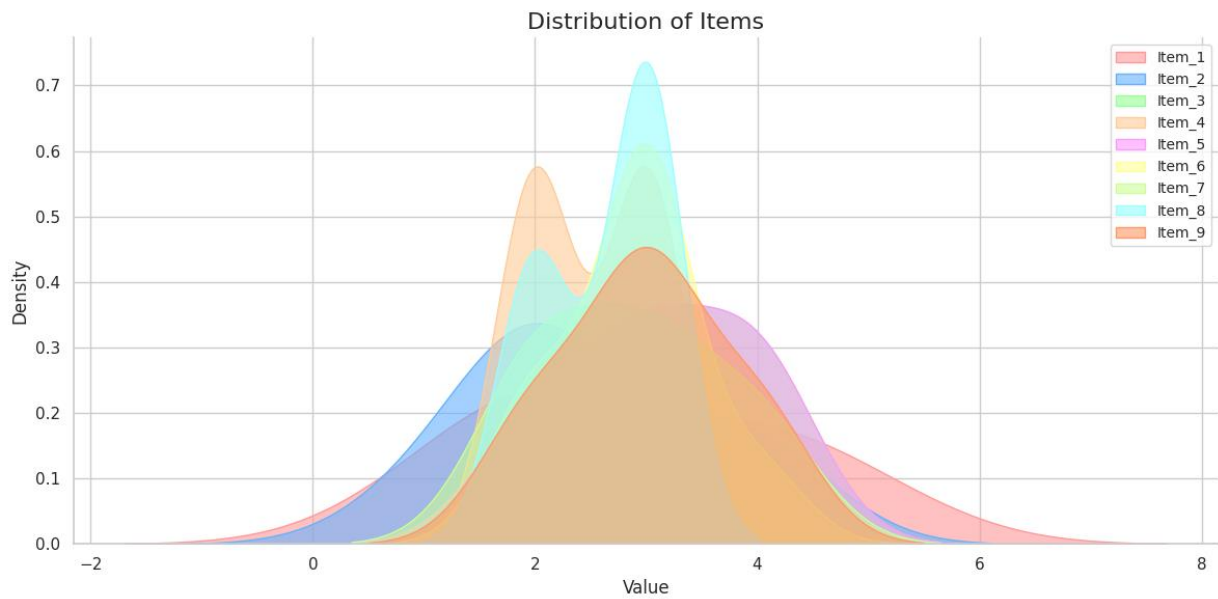


Figure 12 - ADHD items 1-9

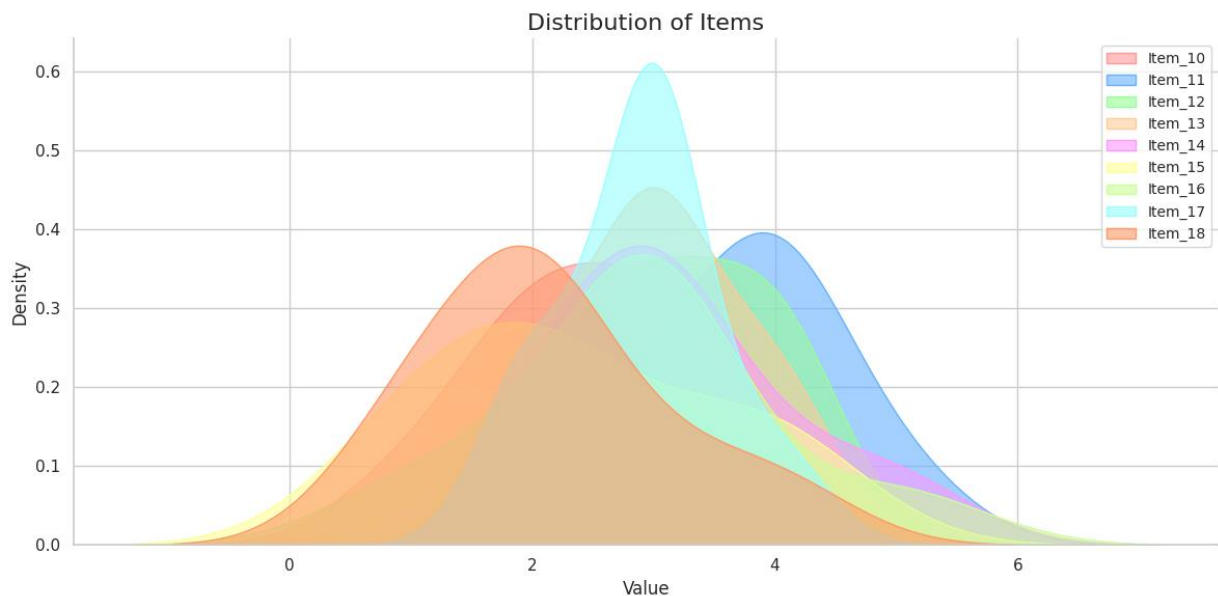


Figure 13 - ADHD items 10-18

The distributions of values across the 18 items show considerable variation, with most items displaying non-uniform distributions across the value range from 1 to 5. Many items (such as 1, 2, and 3) show distinct peaks or concentrations at certain points along the scale. Items like 12, 15, and 18 have broader, more even distributions, while others like 8, 9, and 14 appear to have higher frequencies towards one end of the scale. Unlike the previous description, there's notable diversity in distribution shapes, with few items maintaining consistent frequencies across the entire range.

5.4. Performance Metrics

5.4.1. Performance Metrics - Alexithymia

Accuracy

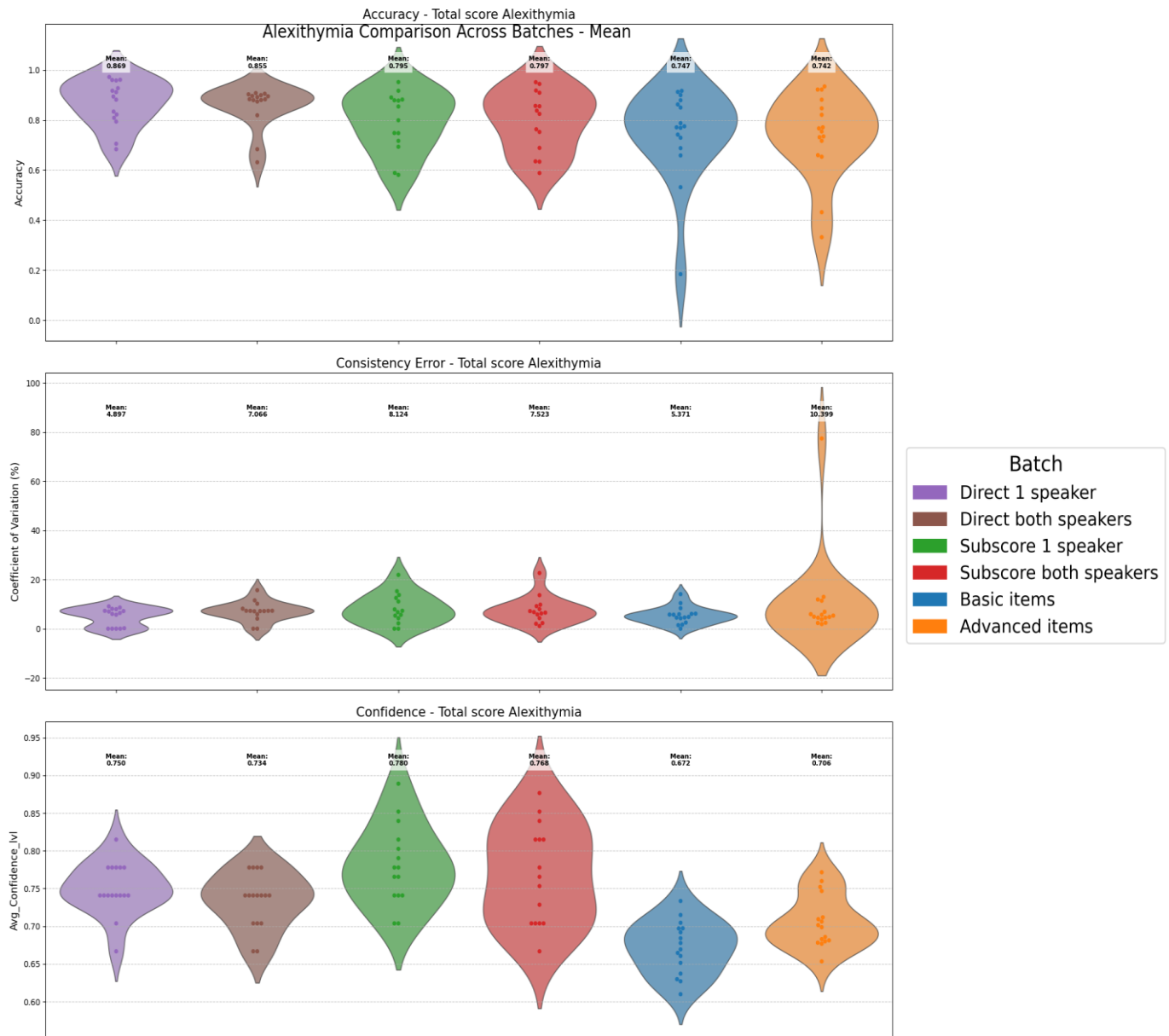


Figure 14 - Total score of Alexithymia across batches (Mean)

Figure 14 presents three violin plots that compare different batches or prompts used in an alexithymia benchmark across three key metrics: accuracy, consistency error, and confidence. The top plot shows accuracy scores, where most batches have mean accuracies ranging between 0.8 and 1.0, with the "Basic items" batch (blue) exhibiting the widest distribution. The middle plot illustrates consistency error, with generally low mean errors across batches, although the "Advanced items" batch (orange) stands out for having the highest and most varied consistency errors. The bottom plot displays confidence scores, where most batches have mean confidence levels between 0.75 and 0.8, with the "Subscale 1 speaker" and "Subscale both speakers" batches (green)

and red) showing the highest mean confidence. Each batch is color-coded, allowing for a clear comparison of performance and variability across these metrics for different prompt strategies or item types in the alexithymia assessment.

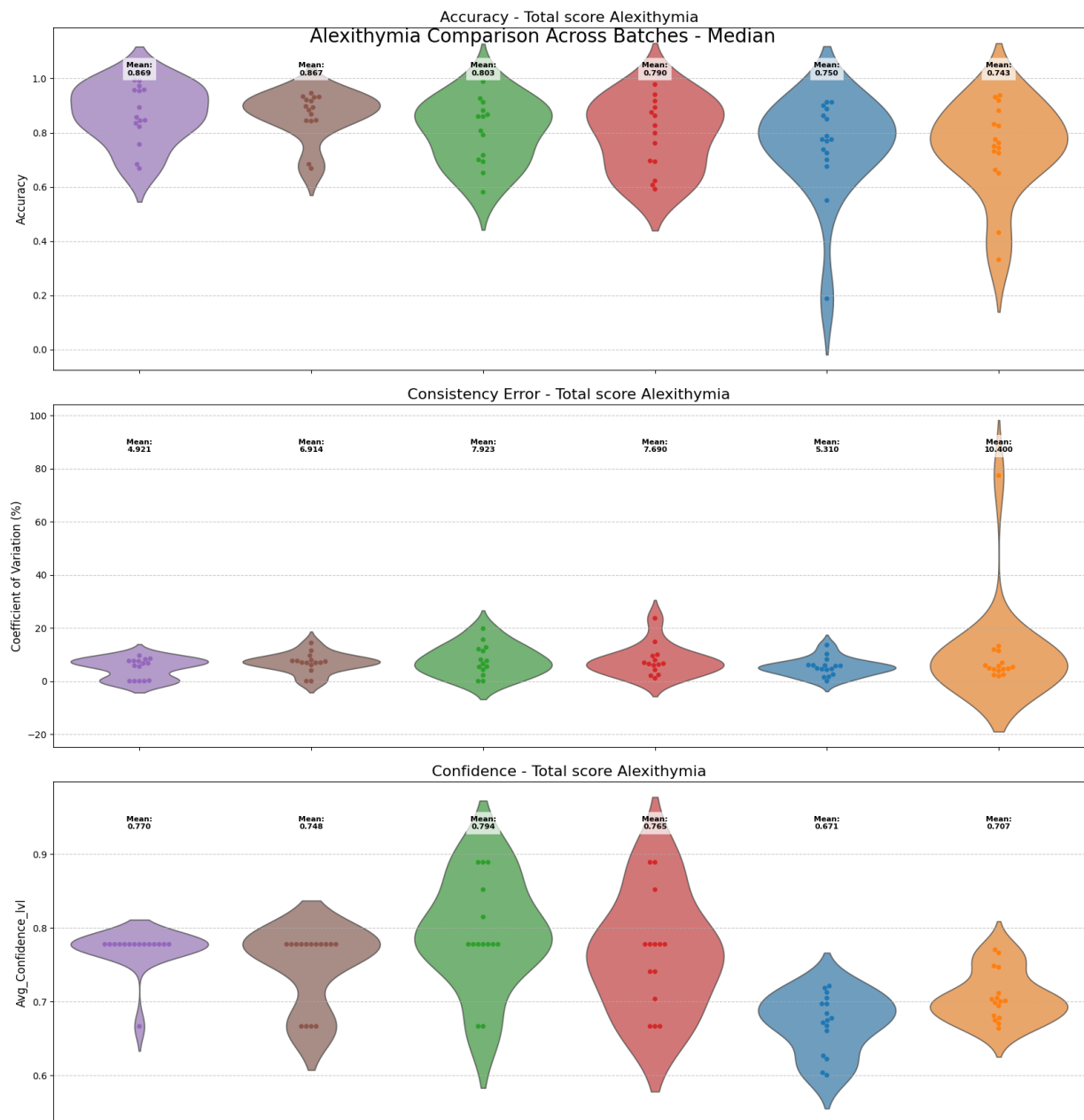


Figure 15 - Total score of Alexithymia across batches (Median)

Using the median instead of the mean in the analysis did not lead to an improvement in accuracy as we can see in figure 15. The distribution of accuracy scores remained similar, with no significant reduction in variability or enhancement in performance. This suggests that the choice of central tendency measure did not have a substantial impact on the overall accuracy outcomes.

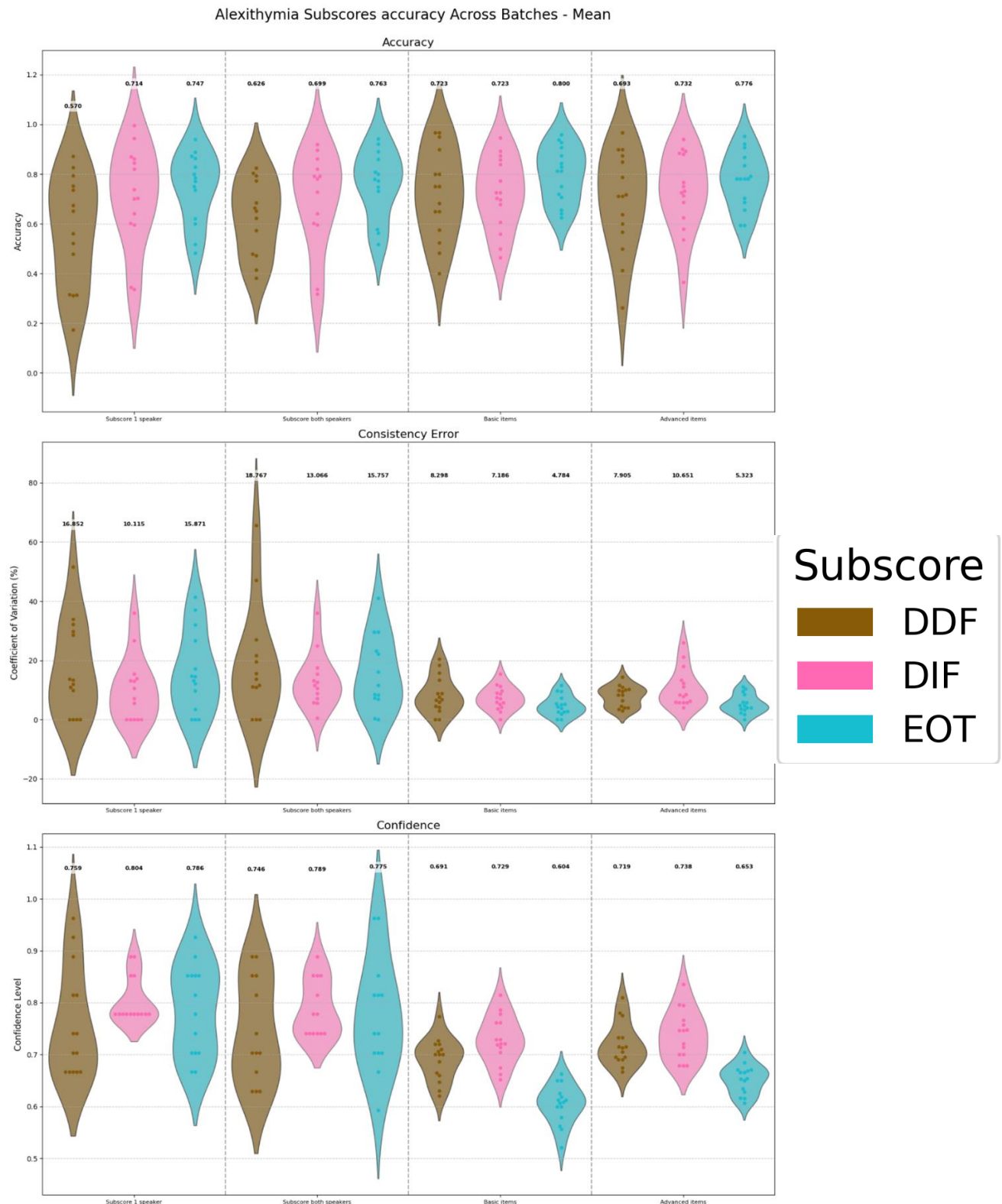


Figure 16 - Alexithymia sub scores across batches (Mean)

Figure 16 presents three violin plots that compare alexithymia subscore accuracies across different batches, focusing on three key metrics: Accuracy, Consistency Error, and Confidence. The top plot illustrates that the "Basic items" batch generally achieves higher accuracy, with mean values approaching 0.8, while the "Subscore 1 speaker" and "Subscore both speakers" batches show a broader distribution, indicating greater

variability. The middle plot examines consistency error, where the "Advanced items" batch demonstrates low errors, with tightly clustered distributions around mean values between 5% and 10%. In contrast, the "Subscore both speakers" batch exhibits higher and more varied consistency errors, suggesting less stable performance. The bottom plot reveals that confidence levels are highest for the "Subscore both speakers" and "Subscore 1 speaker" batches, with mean values around 0.75, whereas the "Basic items" and "Advanced items" batches display slightly lower confidence, with means closer to 0.7. Overall, this figure enables a detailed comparison of how different prompt strategies or item types perform in terms of accuracy, consistency, and confidence within the context of alexithymia assessment.

Alexithymia Subscores accuracy Across Batches - Median

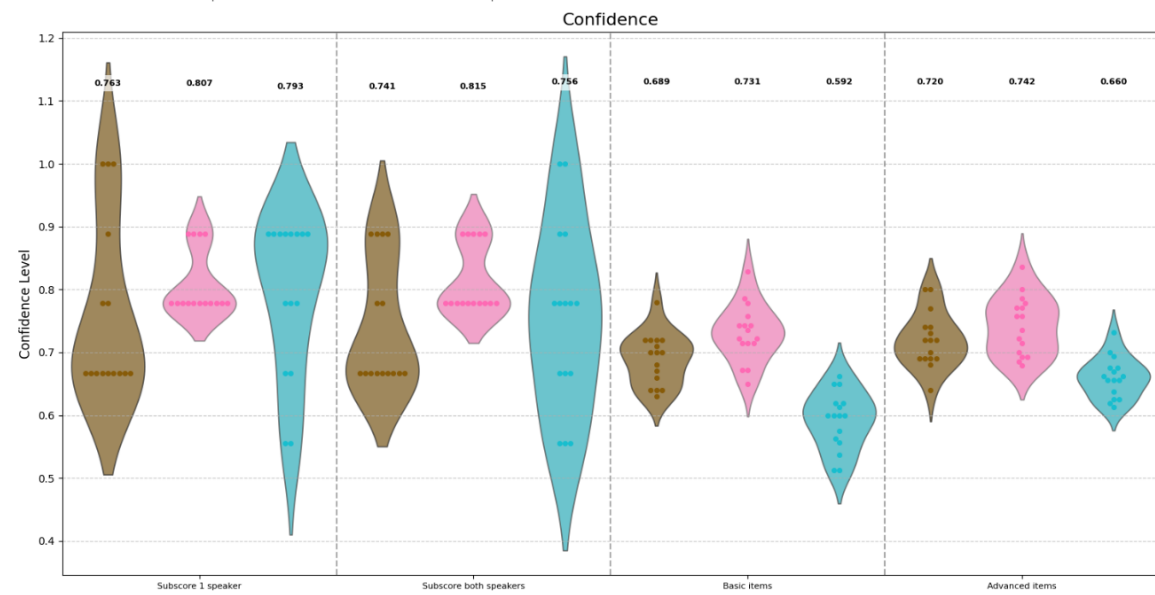
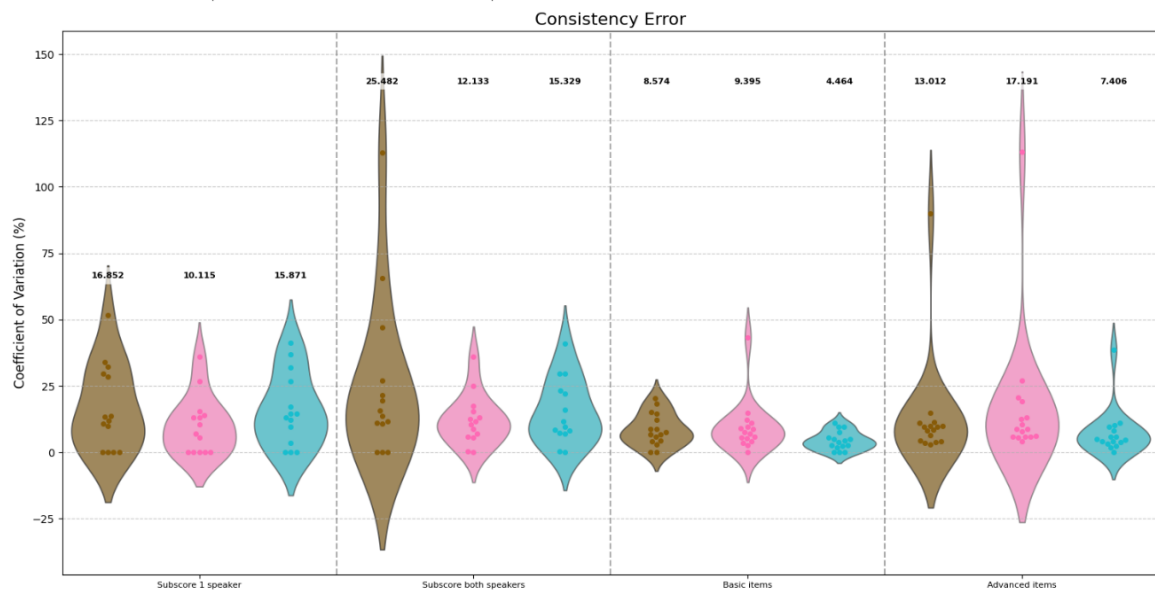
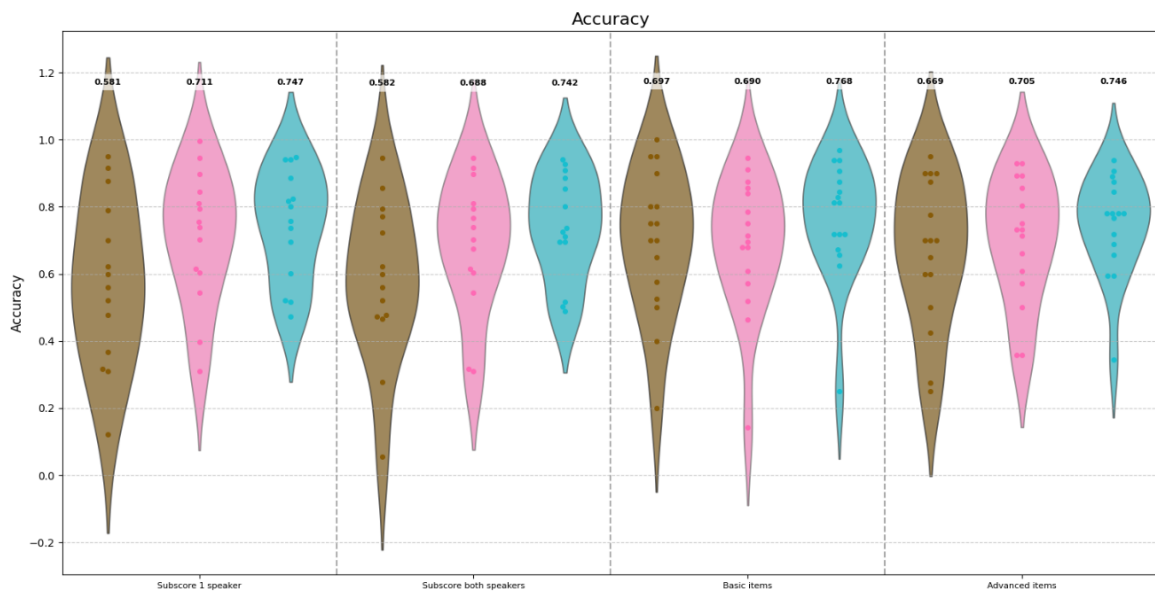


Figure 17 - Alexithymia sub scores across batches (Median)

Comparing the median-based graph (figure 17) to the previous mean-based graph (figure 16), it is evident that using the median did not significantly alter the overall patterns observed in accuracy, consistency error, and confidence. The distributions remain largely similar across the batches, with no marked improvement in accuracy or reduction in variability. This comparison suggests that the choice between median and mean as a central tendency measure has minimal impact on the overall performance and variability of the metrics within the alexithymia assessment.

Predictions vs Target Value

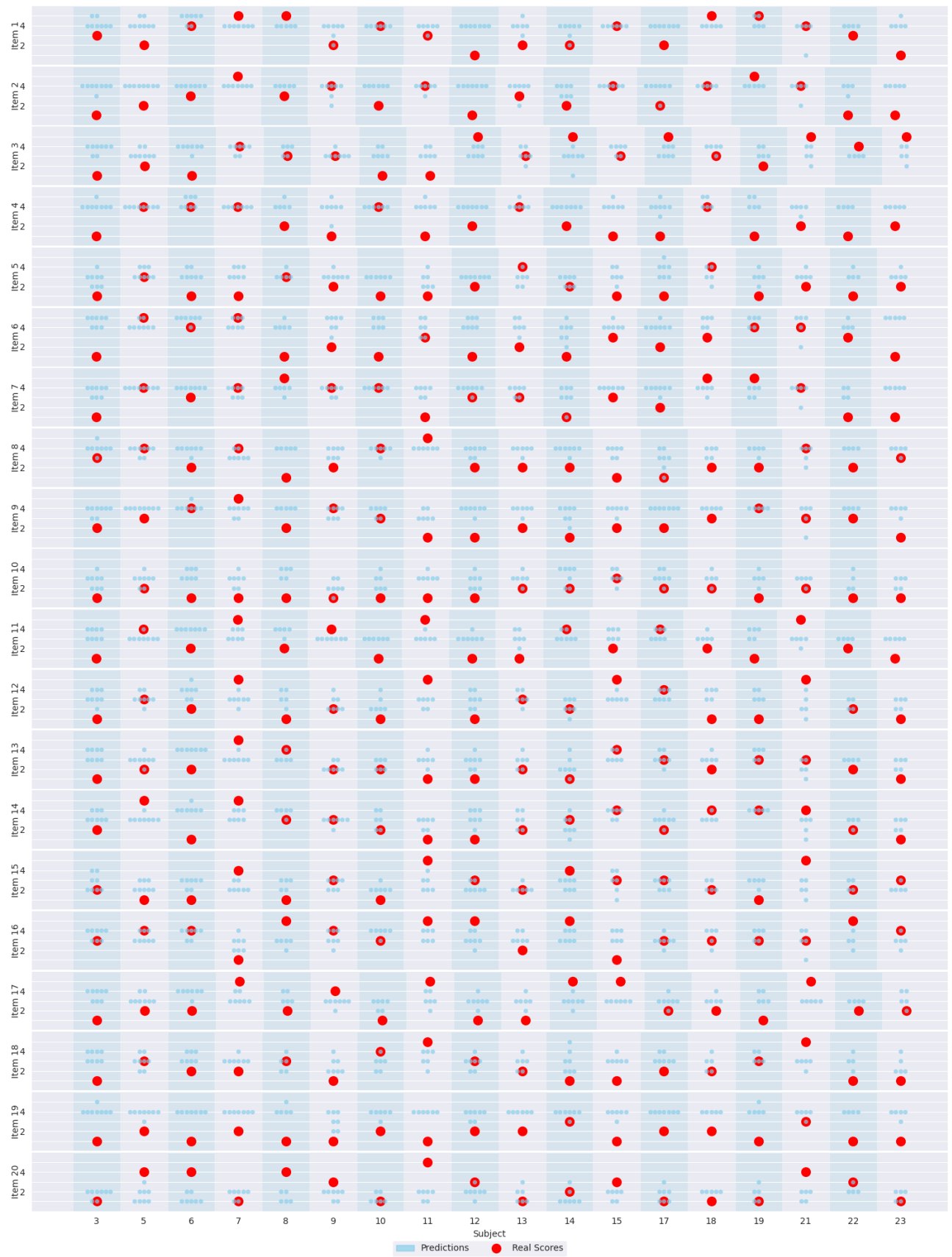


Figure 18 - Predicted scores vs Real Scores - Alexithymia

34

Figure 19 - Item Error Distribution – Alexithymia

Figure 19 presents the aggregated errors per item. If the model overestimates the score, the error would be positive, and the opposite. The general trend is overestimating the scores in most items.

5.4.2. Performance Metrics PTSD

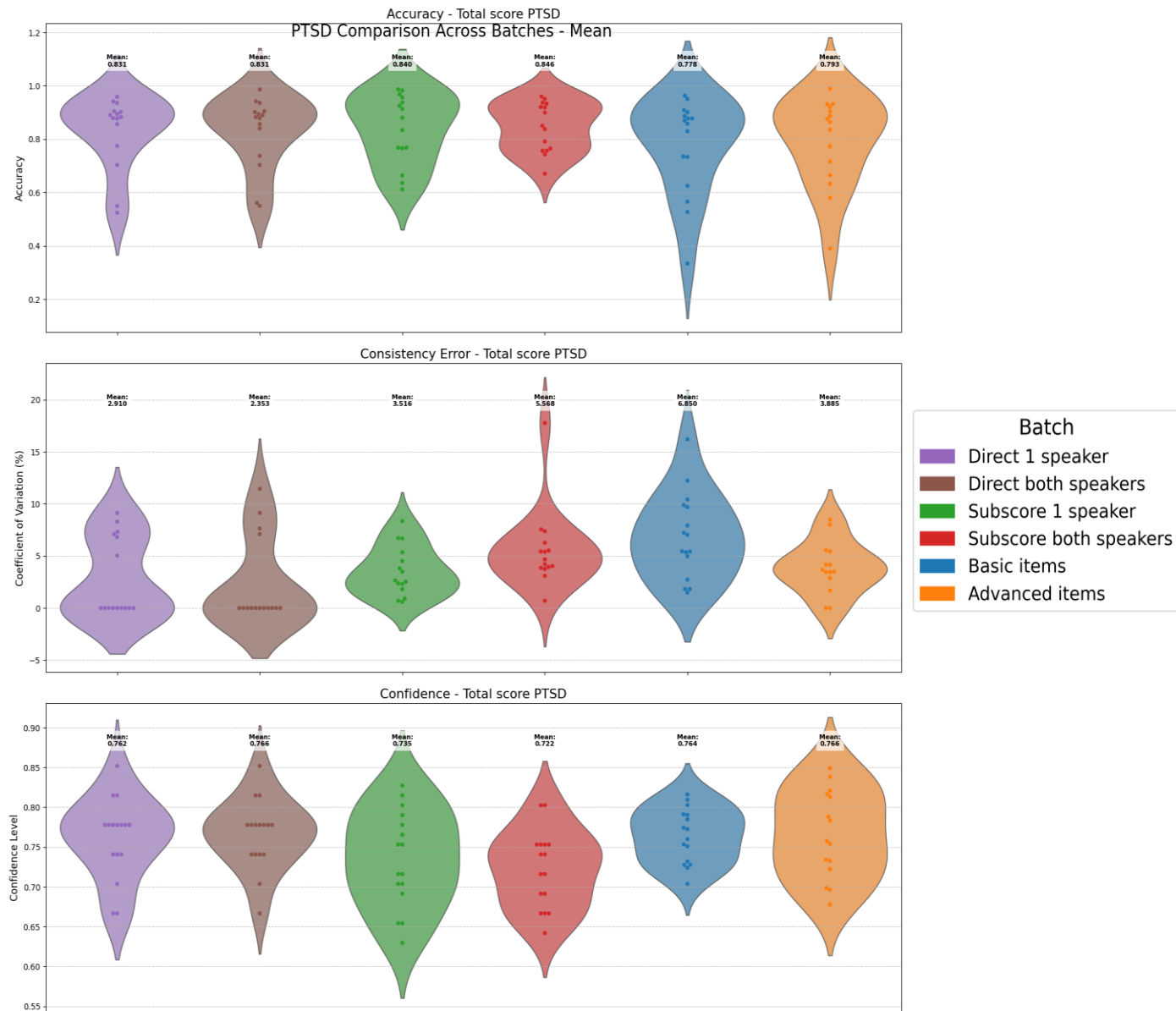


Figure 20 - Total score of PTSD across batches (Mean)

Figure 20 presents a comparison of PTSD assessments across different batches using three metrics: Accuracy, Consistency Error, and Confidence. The data is visualized through violin plots for six distinct batches, each represented by a different color. The top panel shows the accuracy distribution, with mean values ranging from approximately 0.77 to 0.88. The middle panel depicts the consistency error, where most batches exhibit mean errors between 2% and 6%, with two batches showing a notably higher mean error. The bottom panel illustrates the confidence levels, with mean values clustered around 0.72 to 0.76. These plots provide insights into the performance and reliability of PTSD assessments across various conditions or methodologies. Notably, the "Subscore" batches (green and red) demonstrate the highest mean accuracy of 0.84, suggesting it may be the most effective approach for PTSD assessment among the compared methods.

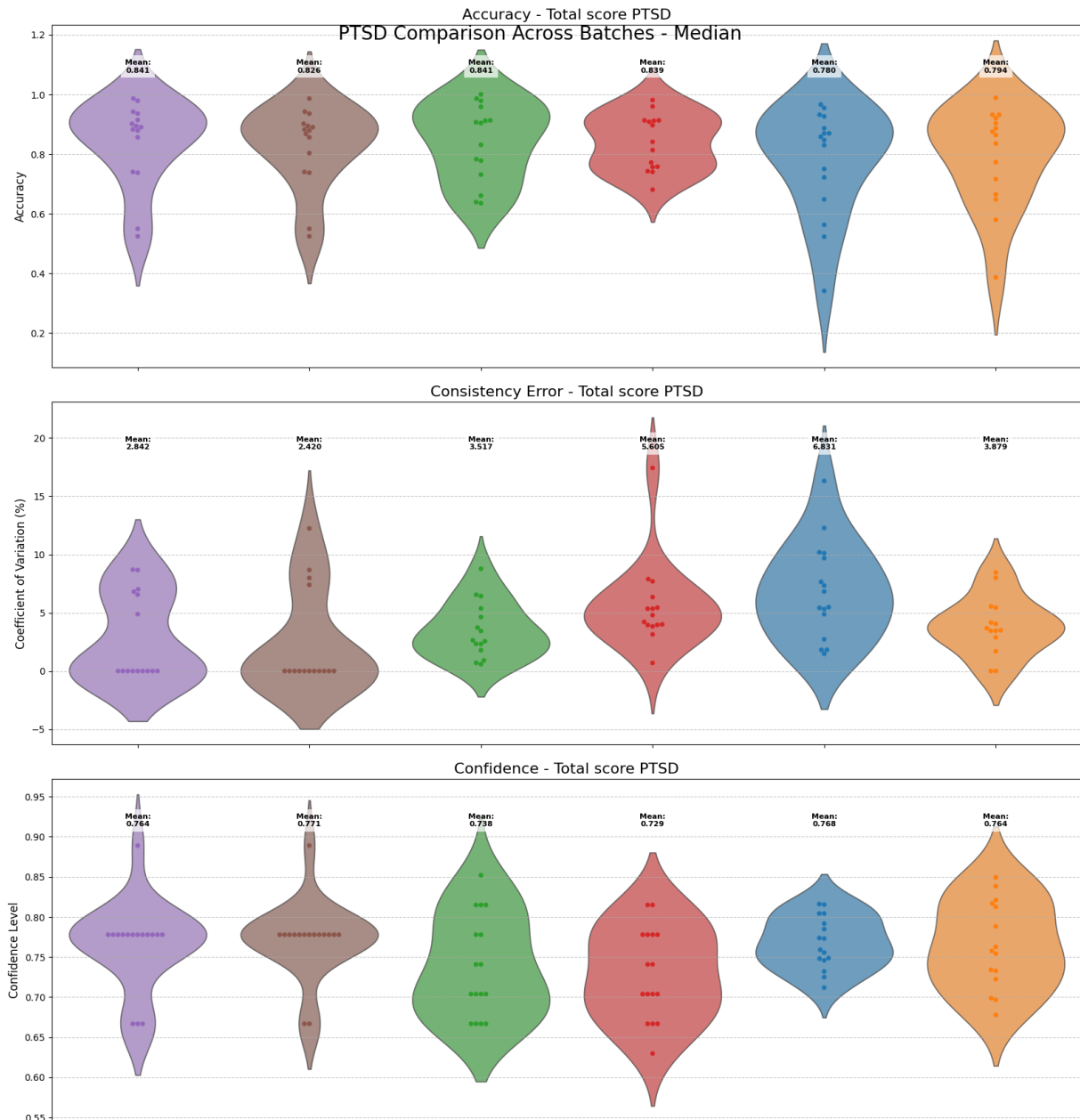


Figure 21 - Total score of PTSD across batches (Median)

If we compare figure 21 to figure 20 we can see that the direct prompts are better in the median figure than in the mean figure, we can explain that by saying that there were “bad runs” of the direct prompt which affected the accuracy. Although the results of the sub score prompt are better in figure 18 we still think the direct approach gives better results.

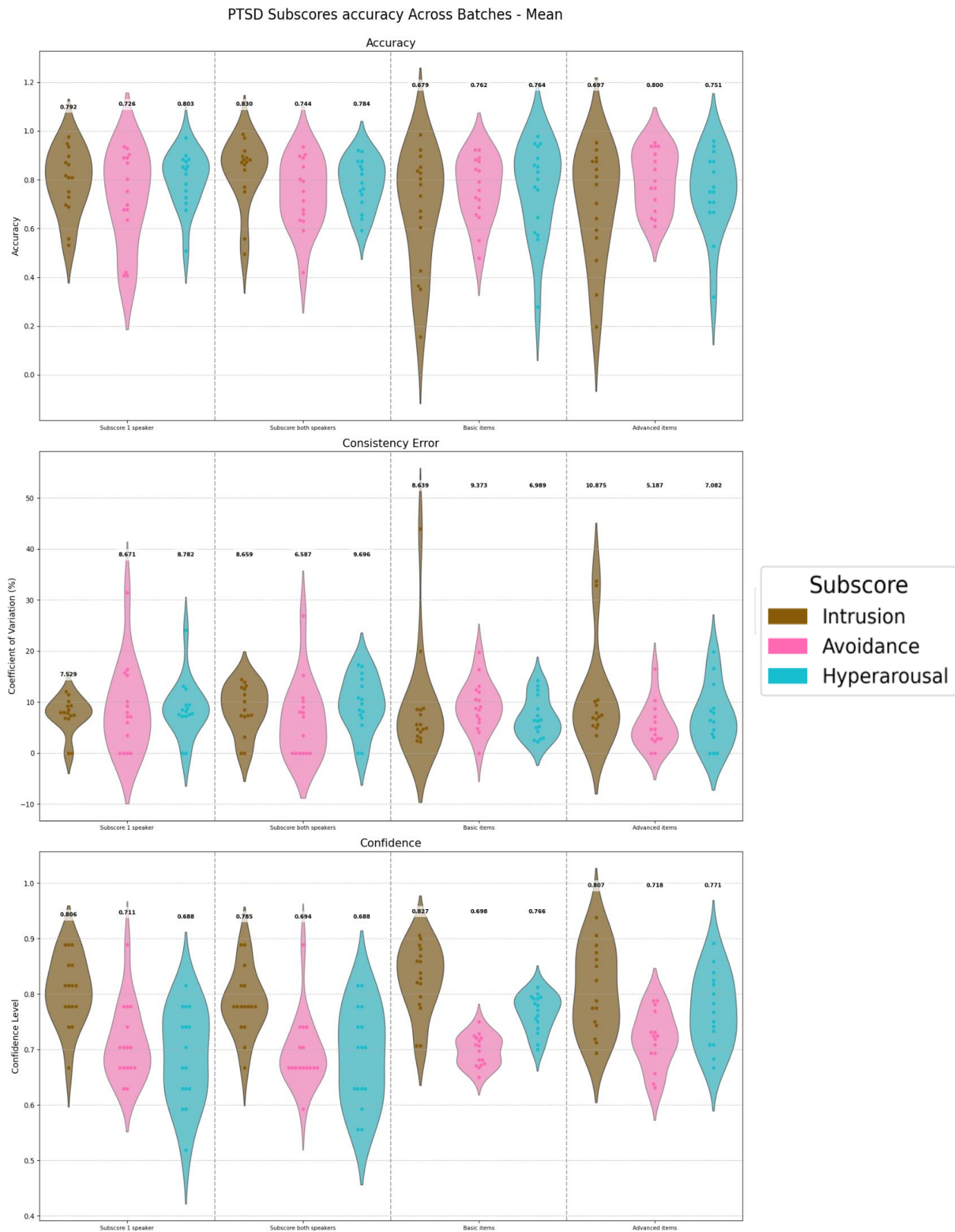


Figure 22 - PTSD sub scores across batches (Mean)

Figure 22 presents three violin plots comparing PTSD sub score accuracies across various batches, with a focus on three key metrics: Accuracy, Consistency Error, and Confidence. The plots reveal that sub score batches generally achieve higher accuracy compared to item batches. Notably, the Intrusion dimension stands out with the highest confidence level, around 80%, despite exhibiting lower accuracy in the item batches. This indicates that while the model is more certain in its predictions for Intrusion, especially at the sub score level, it struggles with accuracy when dealing with individual items within this dimension.

PTSD Subscores accuracy Across Batches - Median

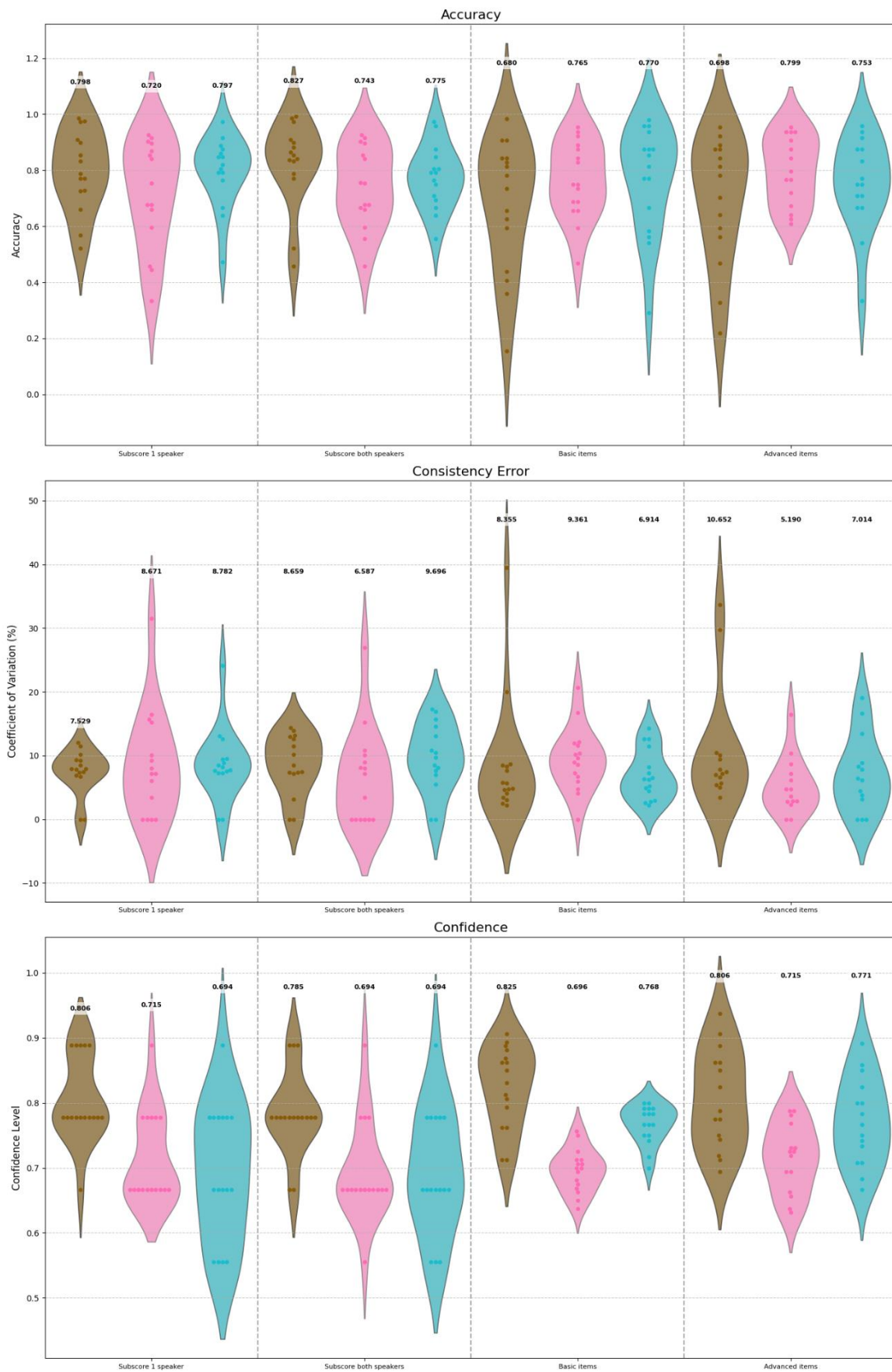


Figure 23 - PTSD sub scores across batches (Median)

Similarly, Figure 23, which presents the median results, shows consistent findings with those observed in Figure 22. The trends in accuracy, consistency error, and confidence across the PTSD sub score batches and item batches remain comparable, reinforcing the observations that sub score batches outperform item batches in accuracy. Additionally, the Intrusion dimension continues to exhibit a high confidence level despite lower accuracy in the item batches, confirming the patterns seen in the mean results.

Predictions vs Target Value

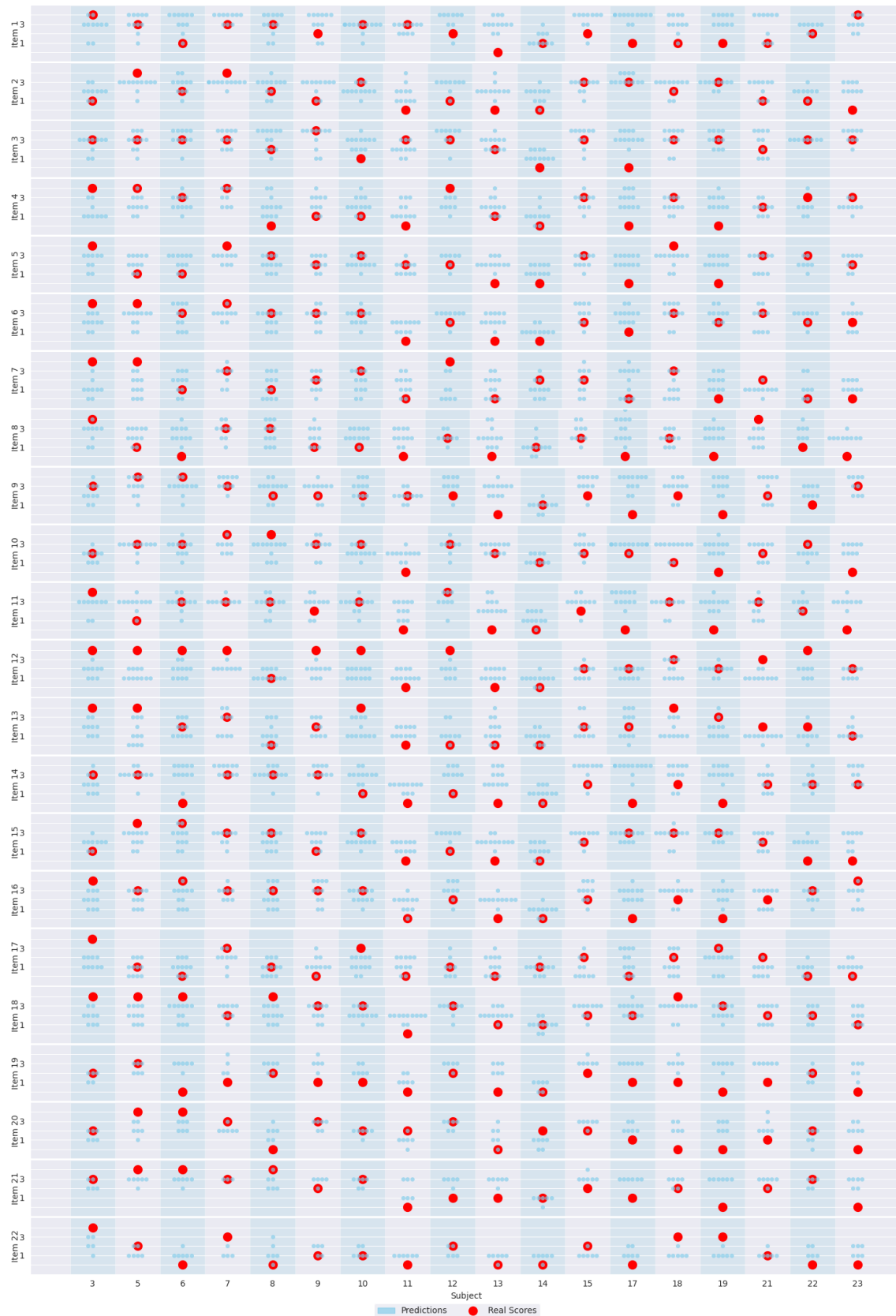


Figure 24 - Predicted scores vs Real Scores - PTSD

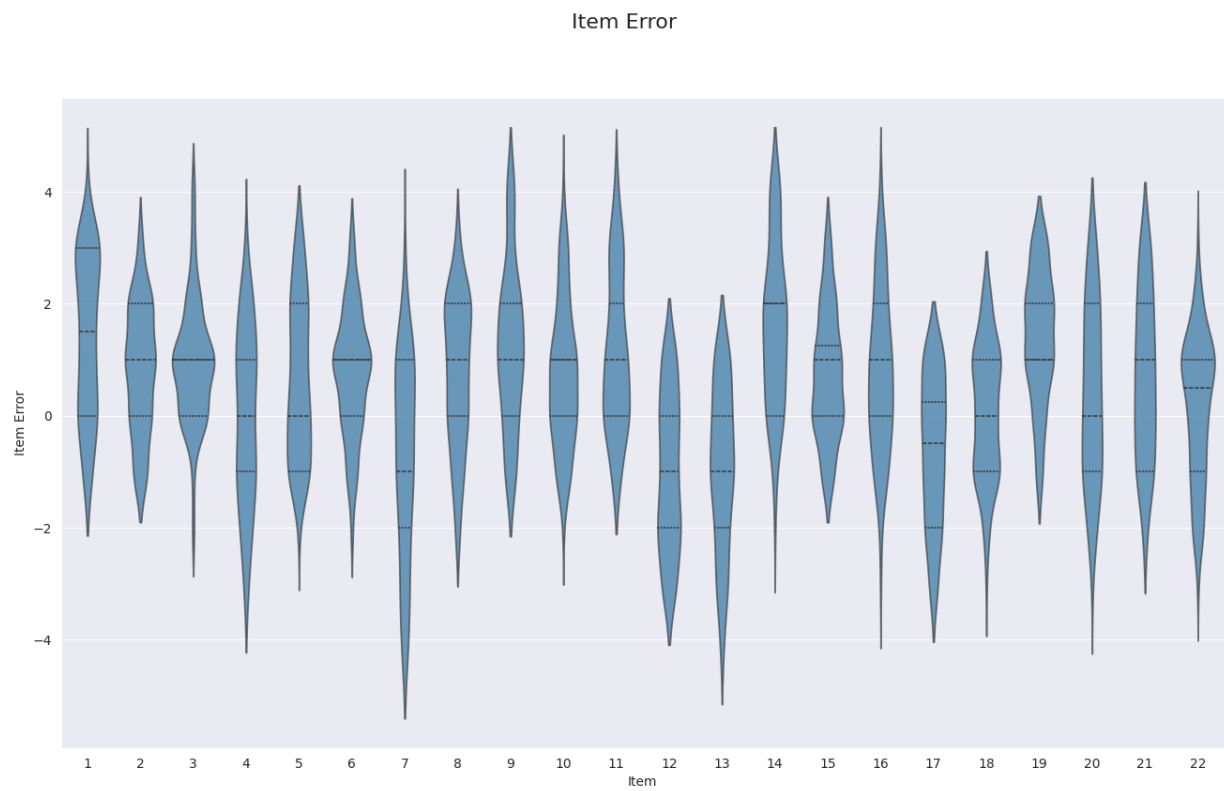


Figure 25 - Item Error Distribution – PTSD

5.4.3. Performance Metrics ADHD

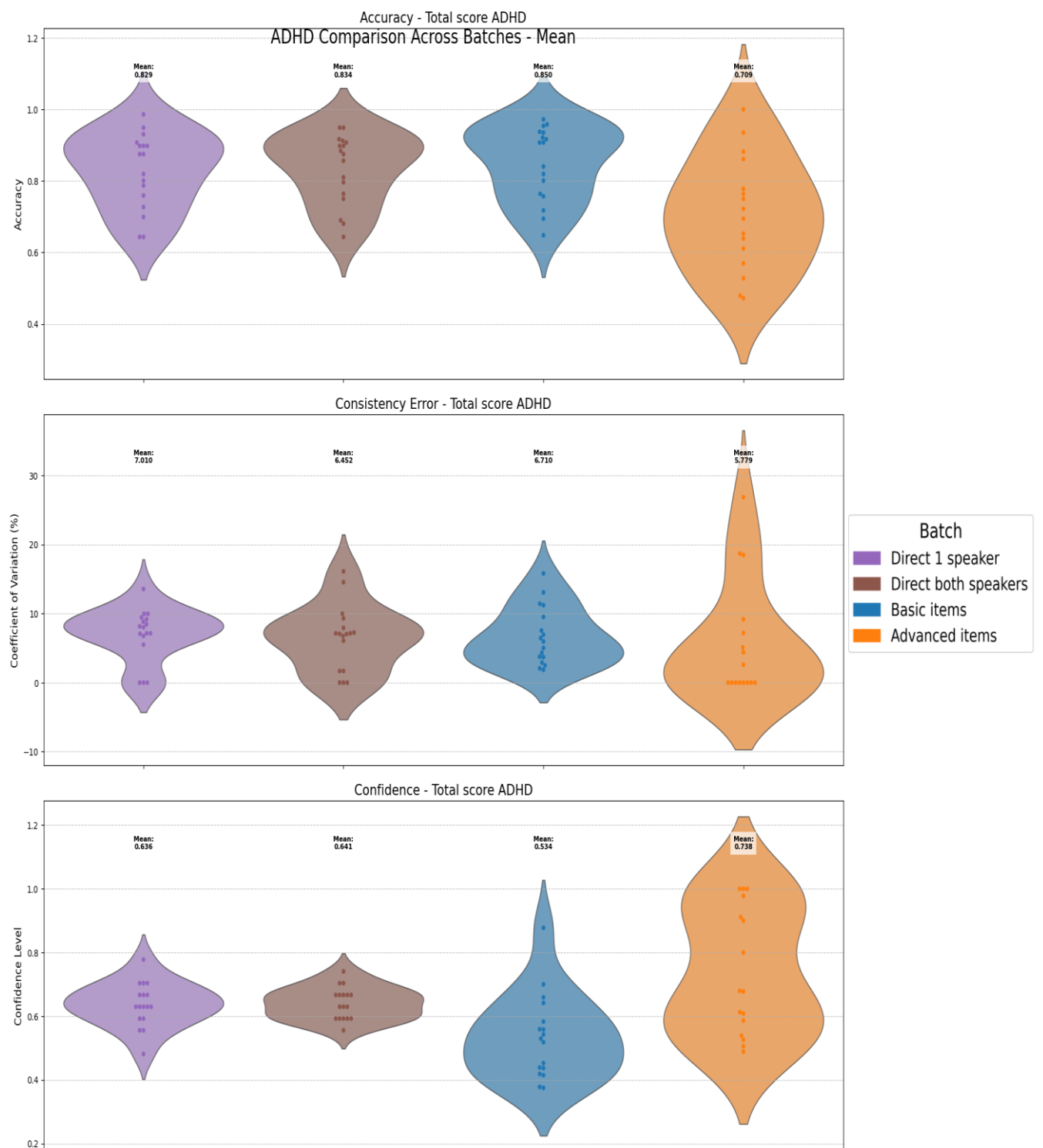


Figure 26 - Total score of ADHD across batches (Mean)

The top panel shows the accuracy distribution. All batches have mean accuracy values around 0.71 to 0.84, with the "Basic items" batch (blue) showing the highest mean accuracy of 0.85.

The middle panel depicts the consistency error. The "Basic items" batch stands out with a much higher mean error (6.71%) and a wider distribution compared to the other batches. The "Advanced items" batch has the lowest error, around 5.8%.

The bottom panel illustrates the confidence levels. Interestingly, we can see all batches have low confidence. We can explain that by mentioning that most of the interviews focus on emotions and feelings and we also analyze the text, while referring to ADHD maybe the right approach will be to analyze the video and the sound of the patient.

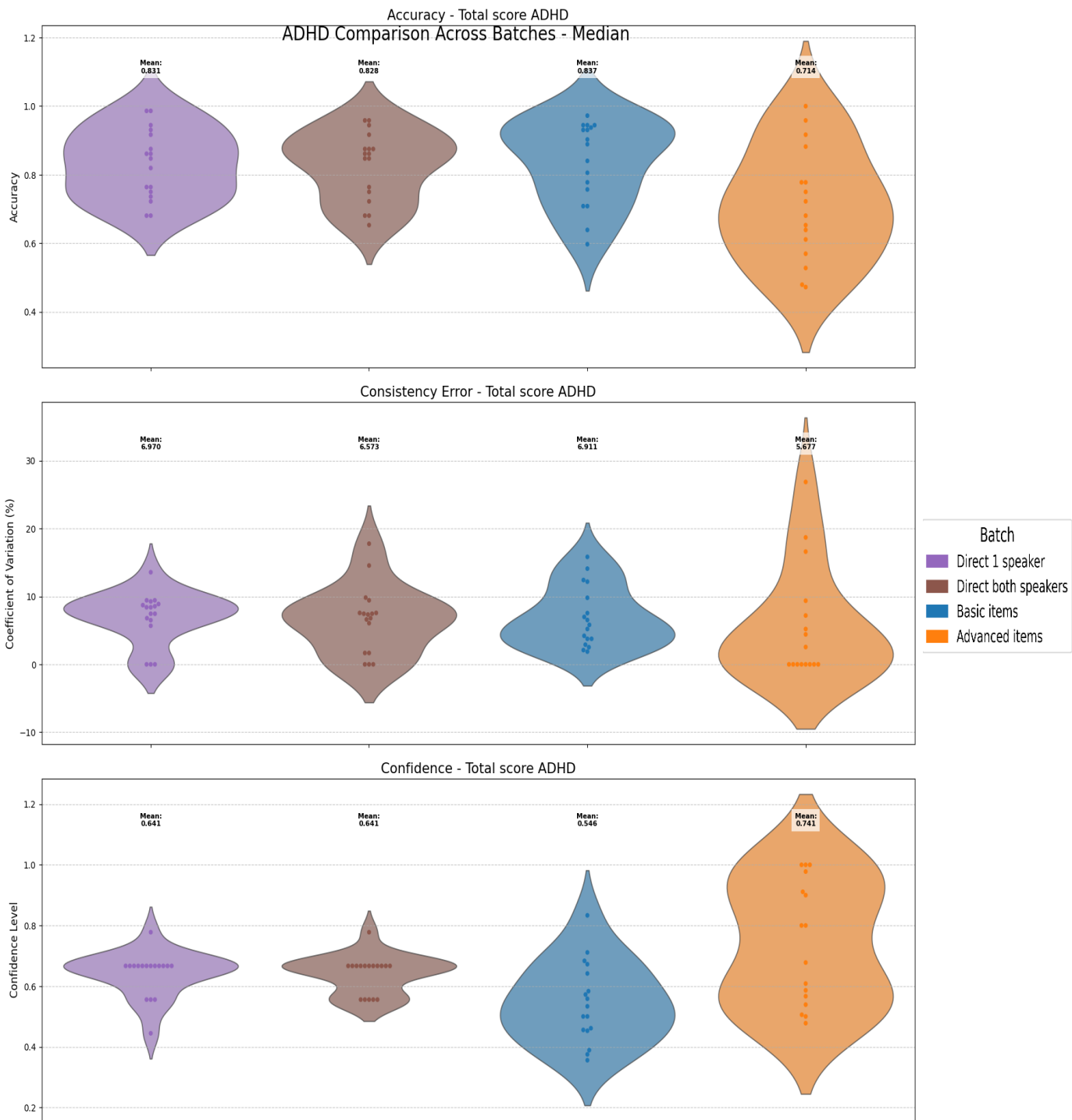


Figure 27 - Total score of ADHD across batches (Median)

In Figure 27 we can see that the batch with the highest accuracy is “Basic items”

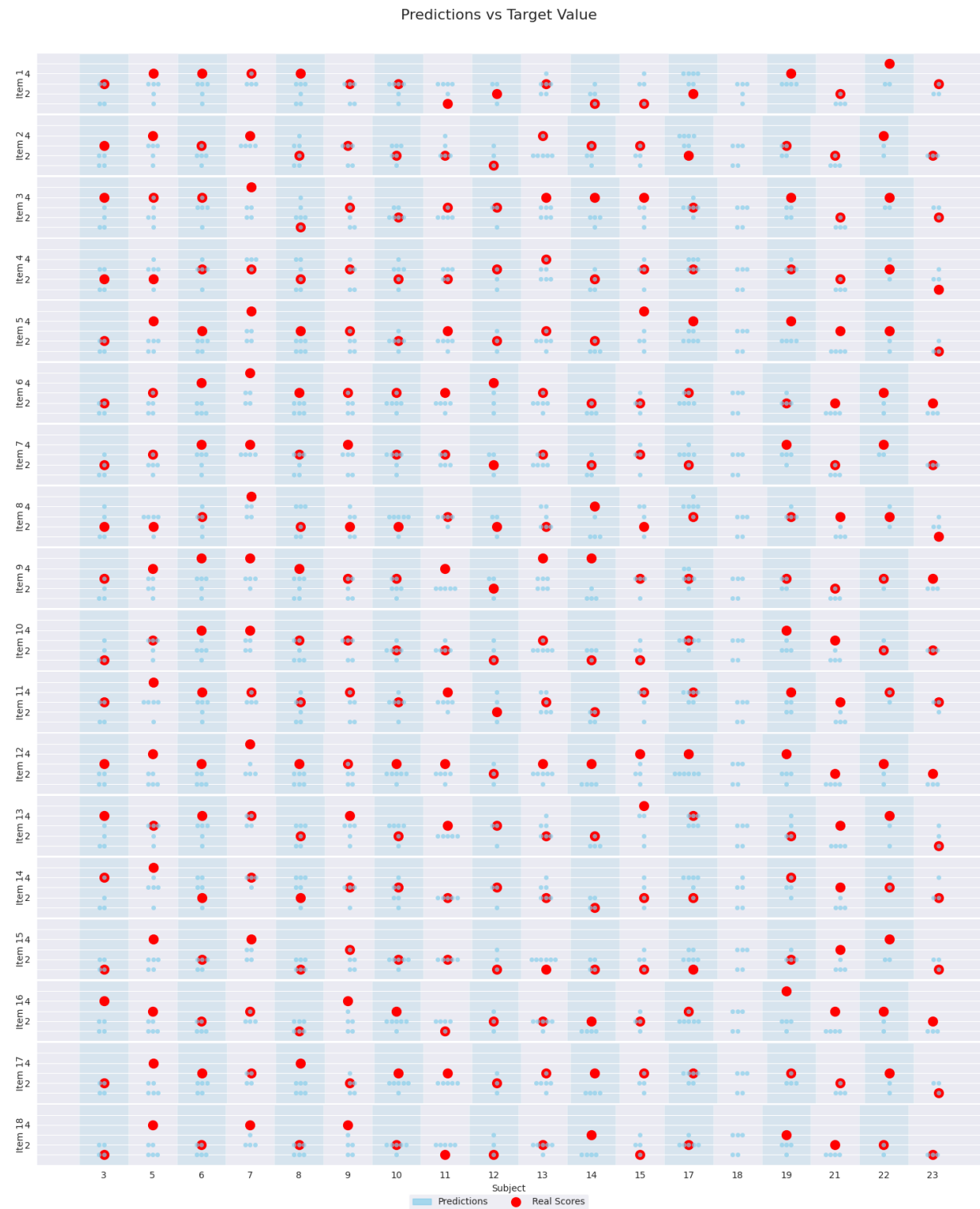


Figure 28 - Predicted scores vs Real Scores - ADHD

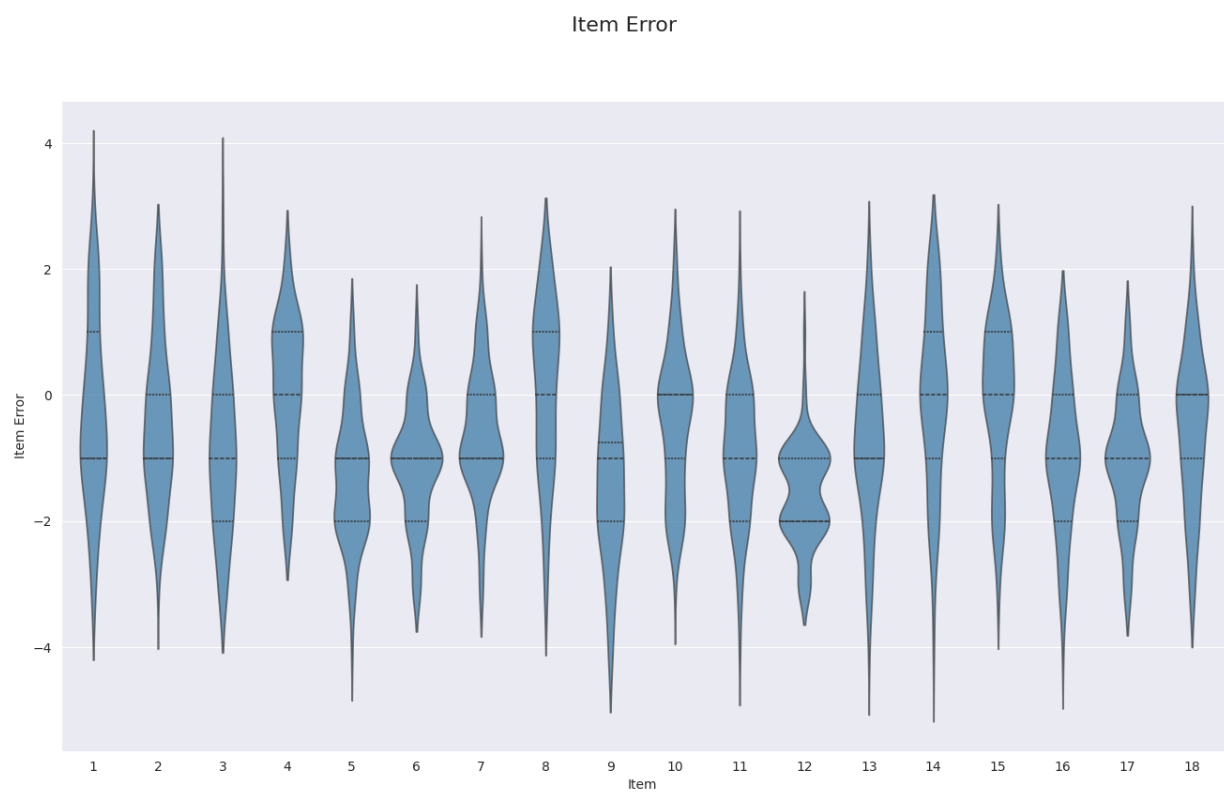


Figure 29 - Item Error Distribution – PTSD

5.5. Statistical analysis

5.5.1. Alexithymia T-Test

In our analysis of the Toronto Alexithymia Scale (TAS-20) responses, we conducted t-tests to compare item-level differences between individuals diagnosed with alexithymia compared to neurotypical. Our hypothesis posited that certain items would demonstrate greater sensitivity in distinguishing between these groups. The results largely supported this hypothesis, with items 2, 11, 12, 17, and 20 showing statistically significant differences ($p < 0.05$) between the two groups.

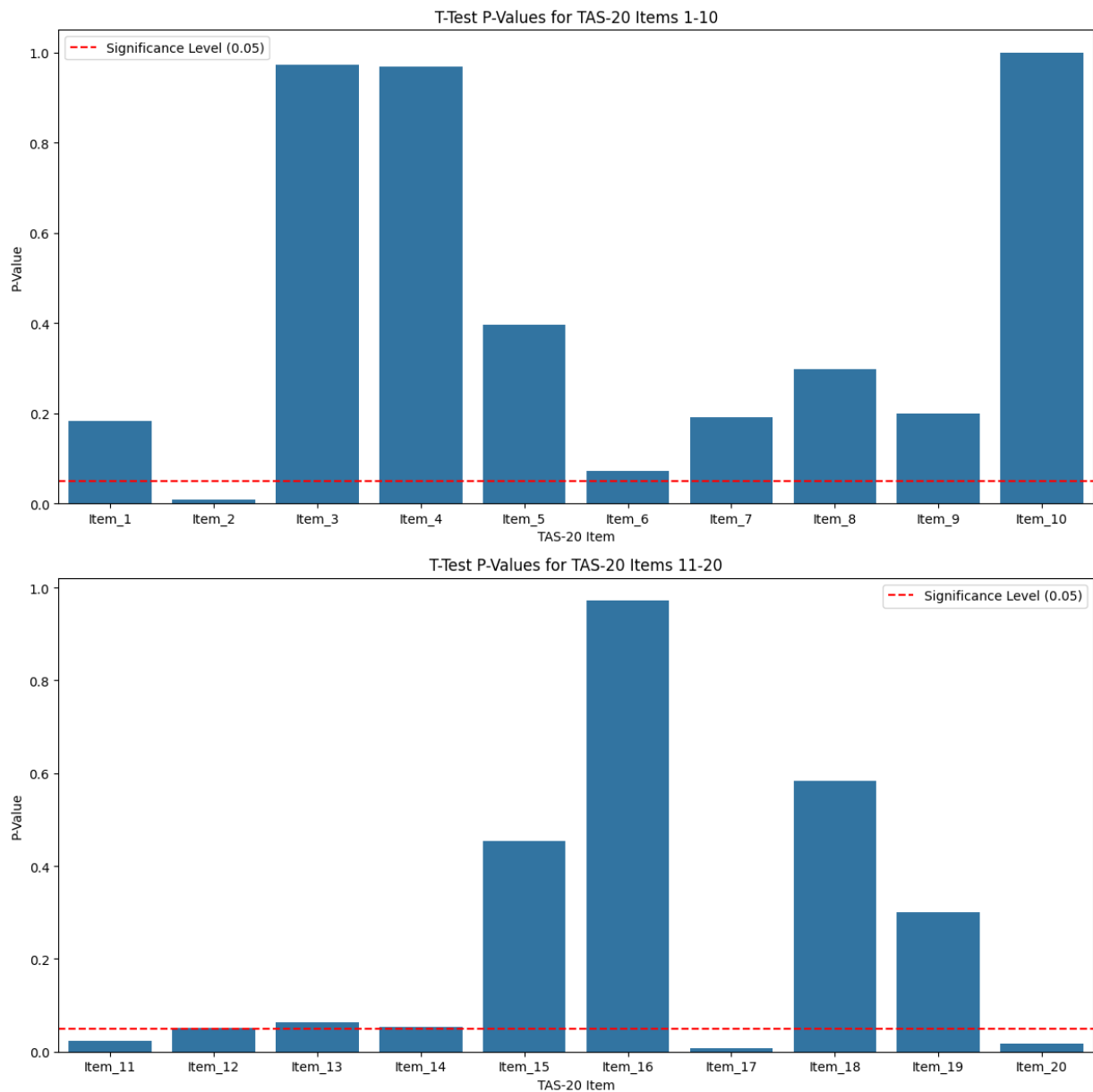


Figure 30 - Alexithymia T-Test

These findings offer valuable insights into the core features of alexithymia that most effectively differentiate affected individuals. Items 2 ("It is difficult for me to find the right words for my feelings"), 11 ("I find it hard to describe how I feel about people"), 12 ("People tell me to describe my feelings more") and 17 ("It is difficult for me to reveal my innermost feelings, even to close friends") all pertain to the difficulty in identifying and expressing emotions, which aligns closely with the fundamental concept of alexithymia. The significance of these items underscores the centrality of emotional articulation challenges in the alexithymia construct.

Interestingly, item 20 ("Looking for hidden meanings in movies or plays distracts from their enjoyment"), which relates to externally-oriented thinking, also emerged as a significant differentiator. This suggests that the tendency towards concrete thinking, often associated with alexithymia, may be a more robust indicator of the condition than previously considered. The significance of this item implies that the cognitive style characteristic of alexithymia extends beyond emotional processing to impact broader aspects of abstract thinking and interpretation.

These results have important implications for both research and clinical practice. From a research perspective, they suggest that future studies might benefit from a more focused examination of these specific aspects of alexithymia. Clinically, these findings could inform the development of more targeted assessment tools and interventions. The identified items might serve as a foundation for a brief screening instrument or guide the focus of therapeutic interventions aimed at improving emotional awareness and expression.

However, it is crucial to interpret these findings with caution. While statistical significance provides valuable insights, it is important to consider effect sizes and clinical significance in tandem. Future research should aim to replicate these findings and explore their practical implications in diverse clinical settings. Additionally, investigating why these particular items are more discriminating could lead to a deeper understanding of the alexithymia construct and its manifestations.

5.5.2. PTSD T-Test

Our analysis of the Impact of Event Scale-Revised (IES-R), a widely used measure for assessing post-traumatic stress symptoms, revealed intriguing results when comparing individuals with and without PTSD diagnoses. T-tests conducted on item-level responses showed that all items, except for items 17 and 22, demonstrated statistically significant differences ($p < 0.05$) between the two groups. This finding suggests that most IES-R items are highly sensitive in distinguishing between individuals with and without PTSD. The extensive differentiation across items underscores the multifaceted nature of PTSD symptomatology and the scale's overall effectiveness in capturing these diverse symptoms. However, the non-significant results for items 17 ("I tried to remove it from my memory ") and 22 ("I tried not to talk about it ") warrant further investigation. These items, both related to avoidance symptoms, may be less discriminating due to the prevalence of avoidance as a coping mechanism even among non-PTSD individuals or potential differences in how these specific avoidance behaviors manifest in PTSD. This outcome highlights the

complexity of avoidance symptoms in PTSD and suggests a need for more nuanced assessment of these particular aspects in future research and clinical evaluations.

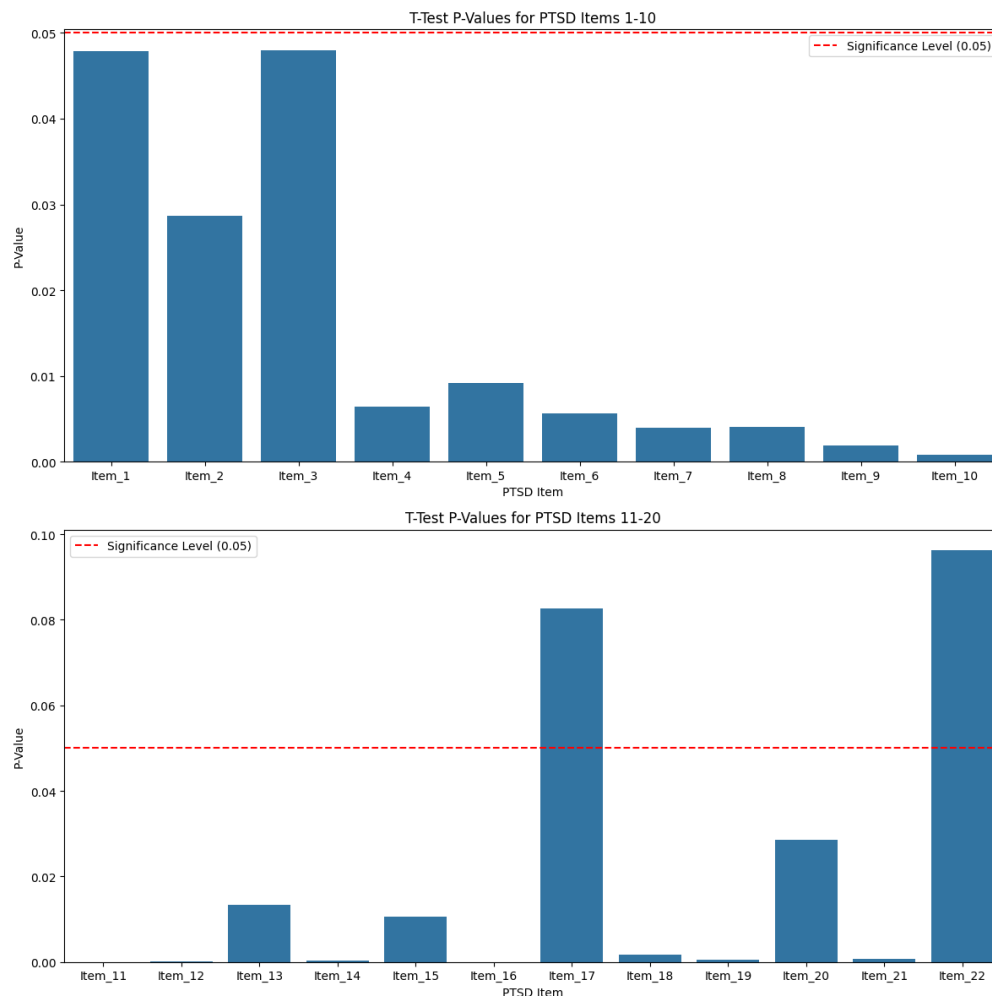


Figure 31 - PTSD T-Test

5.5.3. ADHD T-Test

Our analysis of the Adult ADHD Self-Report Scale-Version 1.1 (ASRS-v1.1), a commonly used tool for screening ADHD symptoms in adults, revealed a more limited degree of differentiation between individuals with and without ADHD diagnoses compared to the IES-R. Independent t-tests conducted on item-level responses indicated that only items 6 ("Often do you feel overly active and compelled to do things, like you were driven by a motor?") and 11 ("Often are you distracted by activity or noise around you ") demonstrated statistically significant differences ($p < 0.05$) between the two groups. This finding suggests that these specific items may be particularly sensitive in distinguishing individuals with ADHD from those without. However, the lack of significant differences for the remaining ASRS-v1.1 items suggests that the scale may have limitations in capturing the full spectrum of ADHD symptoms, particularly those related to inattention and hyperactivity. These results underscore the need for further research to develop more comprehensive assessment tools for ADHD that can effectively differentiate between individuals with and without

the disorder, especially in cases where the presentation of symptoms may be less typical. Additionally, exploring the potential underlying reasons for the differential sensitivity of specific ASRS-v1.1 items may provide valuable insights into the nature of ADHD symptoms and their assessment.

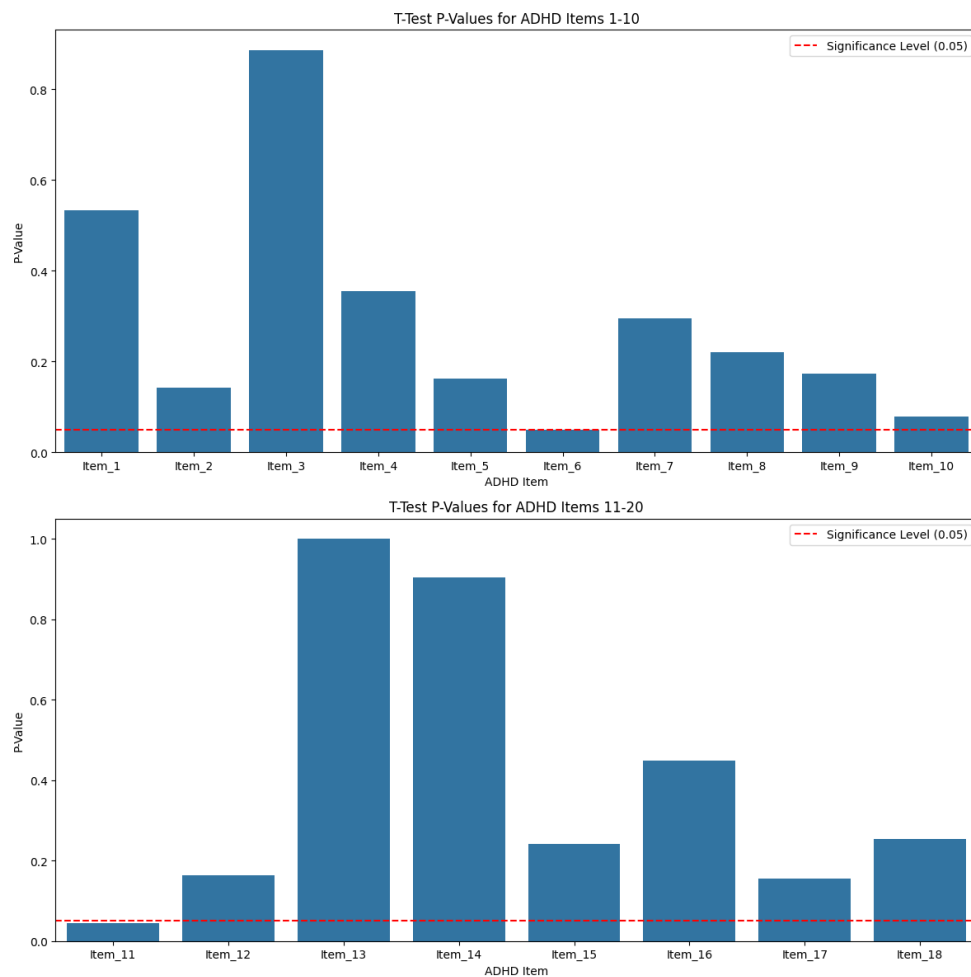


Figure 32 - ADHD - T-Test

5.5.4. Chi-square test

The observed chi-square values for Alexithymia (0.89), PTSD (8.67), and ADHD (2) as we can see in figure 32 were all relatively low, suggesting that the distributions of these scores are not significantly different from a normal distribution. This conclusion is further supported by the p-values of 1.0 for all three conditions, indicating that there is no strong evidence to reject the null hypothesis of normality for any of them.

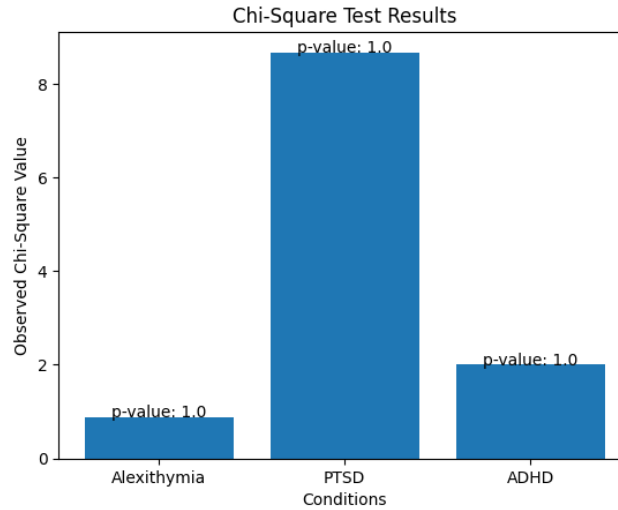


Figure 33 - Chi-square test

5.5.5. Power Analysis

The power of a statistical test is a crucial consideration in research design, as it determines the likelihood of detecting an effect when there is one. In our study, we evaluated the required sample sizes to achieve 80% power for assessing the severity of Alexithymia, PTSD, and ADHD using established effect sizes from prior analyses.

To conduct the power analysis, we utilized the following method:

Effect Sizes:

The effect sizes used for each condition were derived from our data using a custom calculation method. This method calculates an effect size similar to Cohen's d, but adapted for our specific dataset structure: a. For each subject, we calculated the mean score across all items in the questionnaire. b. We then calculated the standard deviation of these scores for each subject. c. The overall mean across all subjects' mean scores was computed. d. The average of all subjects' standard deviations was calculated. e. The effect size was then computed as:

$$Effect\ Size = \frac{\mu_{subjects}}{\bar{\sigma}_{subjects}} \quad Eq. (5.5.3.1):$$

Where:

- $\mu_{subjects}$ is the overall mean of subjects' mean scores
- $\bar{\sigma}_{subjects}$ is the average of subjects' standard deviations

This method provides a measure of the strength of the effect relative to the variability within the data. Using this approach, we calculated the following effect sizes:

- Alexithymia: 1.1922
- PTSD: 0.9198
- ADHD: 1.2101

Standard Deviations:

The standard deviations for the scores from the questionnaires were as follows:

- Alexithymia: 2.4698
- PTSD: 2.4215
- ADHD: 2.9211

Power Analysis Method:

Using these effect sizes and standard deviations, we applied the power analysis to determine the sample sizes needed to reach an 80% power threshold for each condition. The analysis was conducted using the `TTestIndPower` class from the `statsmodels` library, which facilitates the computation of power for varying sample sizes. Prior to conducting the power analysis, we performed a chi-square test for normality on our data. The results yielded p-values of 1.0 for all three conditions, indicating that there is no strong evidence to reject the null hypothesis of normality for any of them. Based on these results, we proceeded with the assumption of Gaussian distribution for our power analysis.

The equations implemented in the power analysis are derived from the following general formula for calculating power ($1 - \beta$) in a two-sample t-test:

$$1 - \beta = \Phi\left(\frac{(d \cdot \sqrt{n})}{\sigma} - z_{\frac{\alpha}{2}}\right) \quad \text{Eq. (5.5.3.2):}$$

where:

- $1 - \beta$ is the power of the test.
- d is the effect size (Cohen's d).
- n is the sample size.
- σ is the standard deviation.
- $z_{\frac{\alpha}{2}}$ is the critical value of the standard normal distribution at significance level α .
- Φ is the cumulative distribution function of the standard normal distribution.

The results of the power analysis indicated the following required sample sizes to achieve the desired statistical power:

- Sample size for Alexithymia to reach 0.8 power: 13 participants
- Sample size for PTSD to reach 0.8 power: 20 participants
- Sample size for ADHD to reach 0.8 power: 12 participants

These findings suggest that the current sample of 20 participants is sufficient for PTSD and ADHD assessments, but additional participants may be needed for Alexithymia to achieve adequate power in the study.

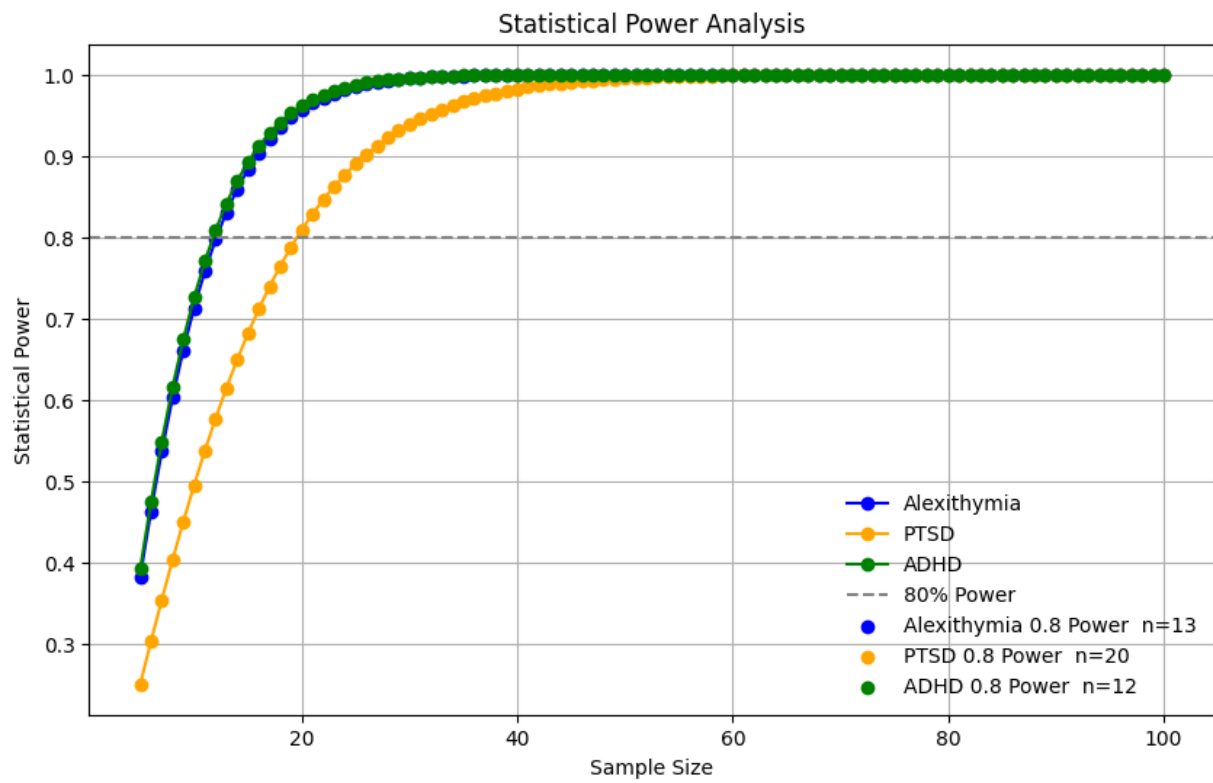


Figure 34 - Power Analysis

5.6. Correlation Between Different Mental States

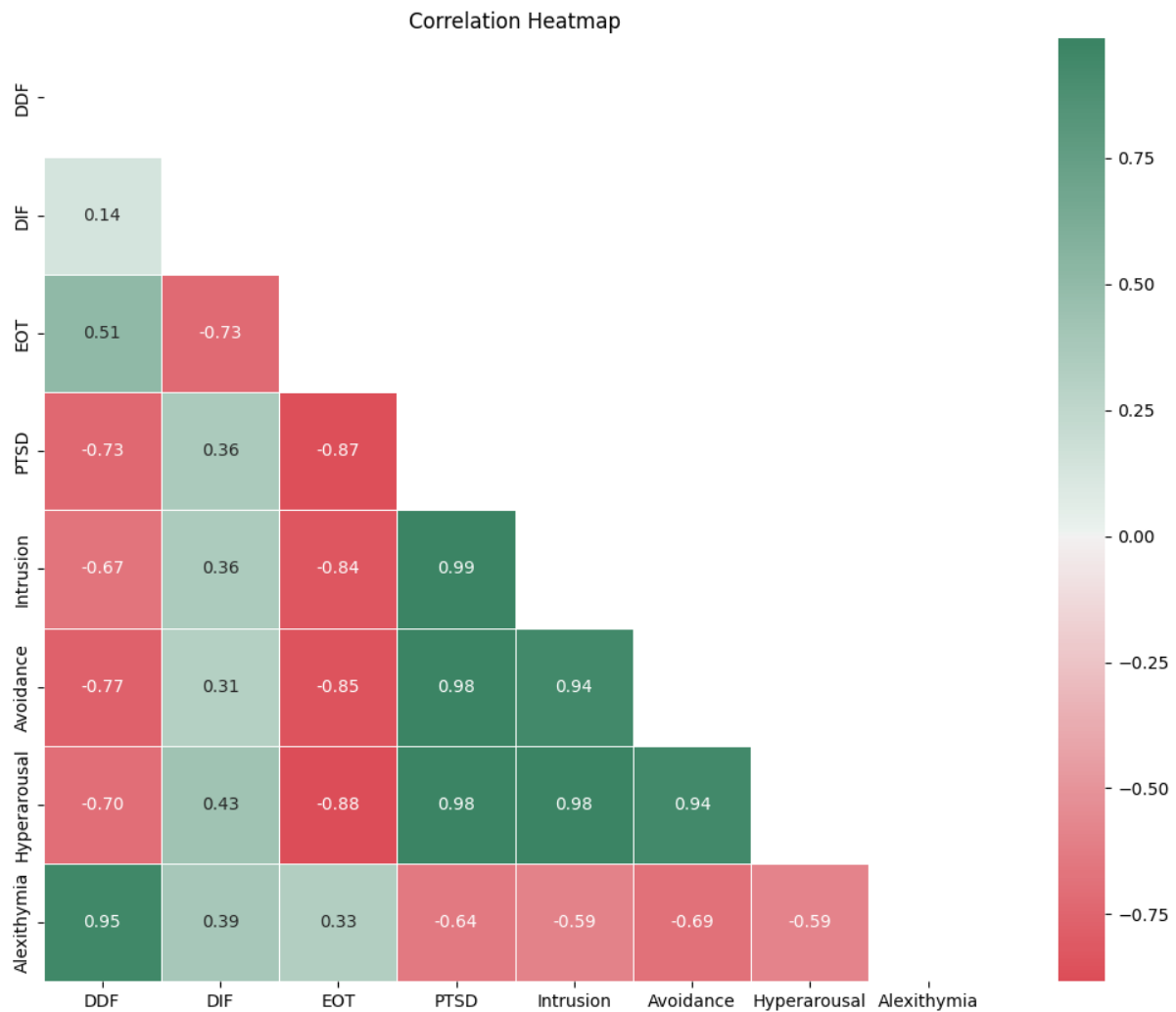


Figure 35 - correlation between Alexithymia symptoms and PTSD symptoms

In figure 35 we see correlation between Alexithymia symptoms and PTSD symptoms. We can see that there is strong inner correlation within each condition's symptoms but seems like an inverse correlation between the severity of Alexithymia and PTSD, I.E. Alexithymic patients are less likely to suffer from PTSD.

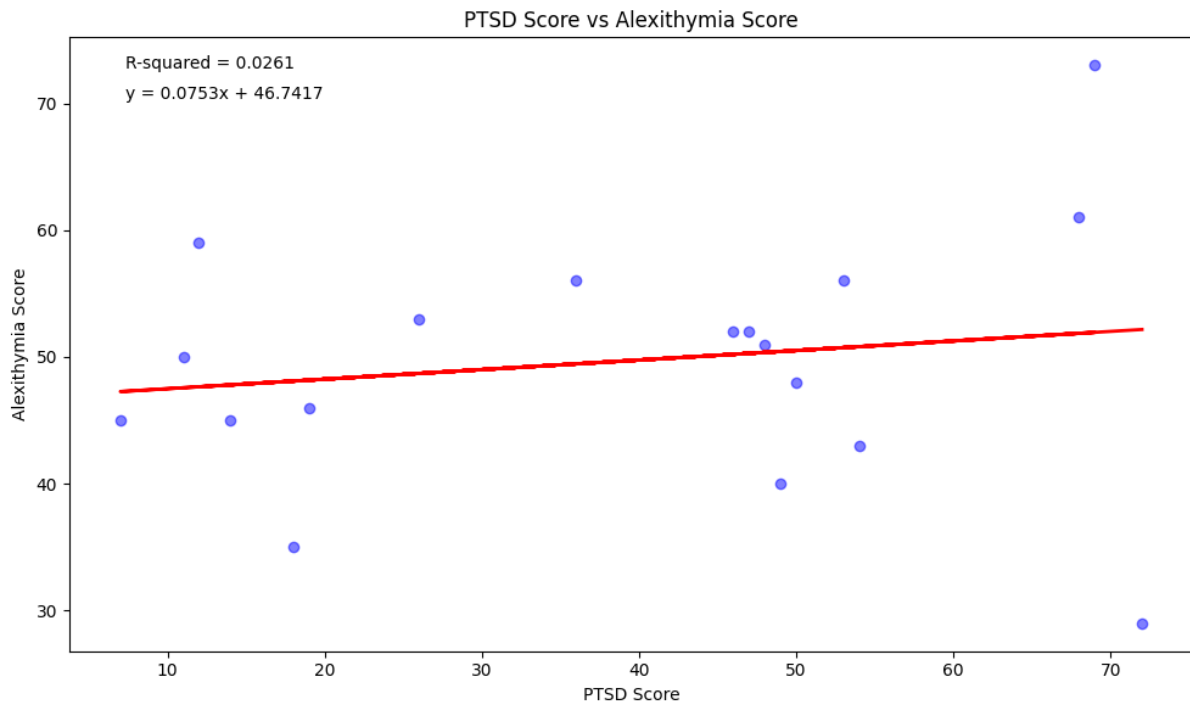


Figure 36 - Linear Regression of PTSD to Alexithymia

In figure 36, we attempted linearly regress the Alexithymia score from the PTSD score. The coefficient is very weak over the general population of the study.

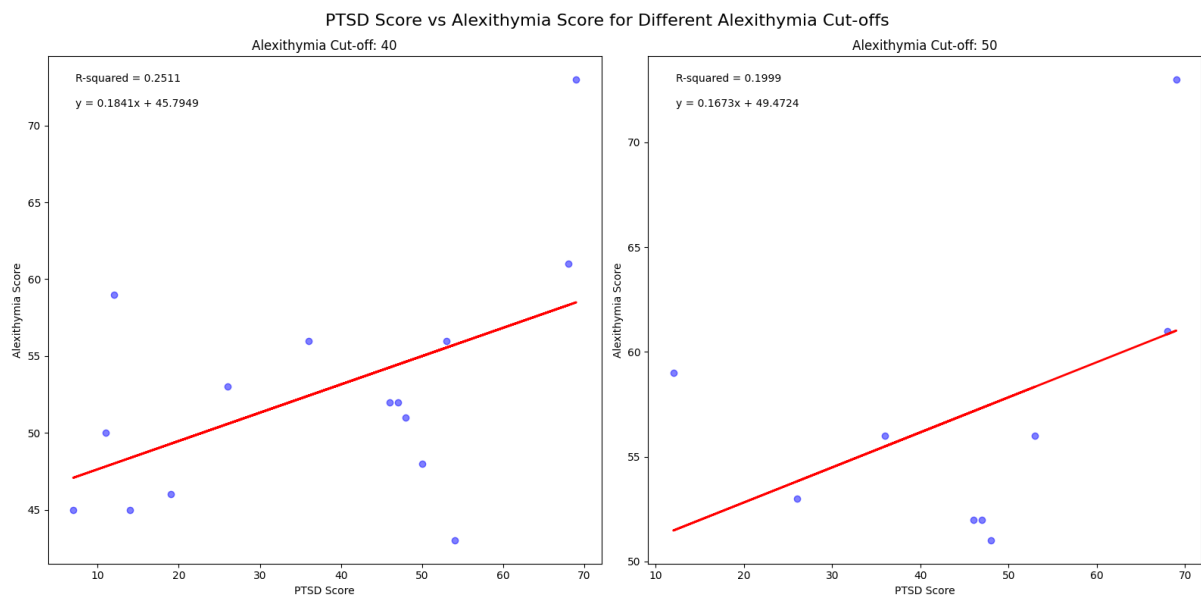


Figure 37 - Linear Regression with cutoffs

In figure 37, we checked if the correlation is stronger in severe cases of PTSD. The coefficient seems to be stronger to some degree, but still not profound enough.

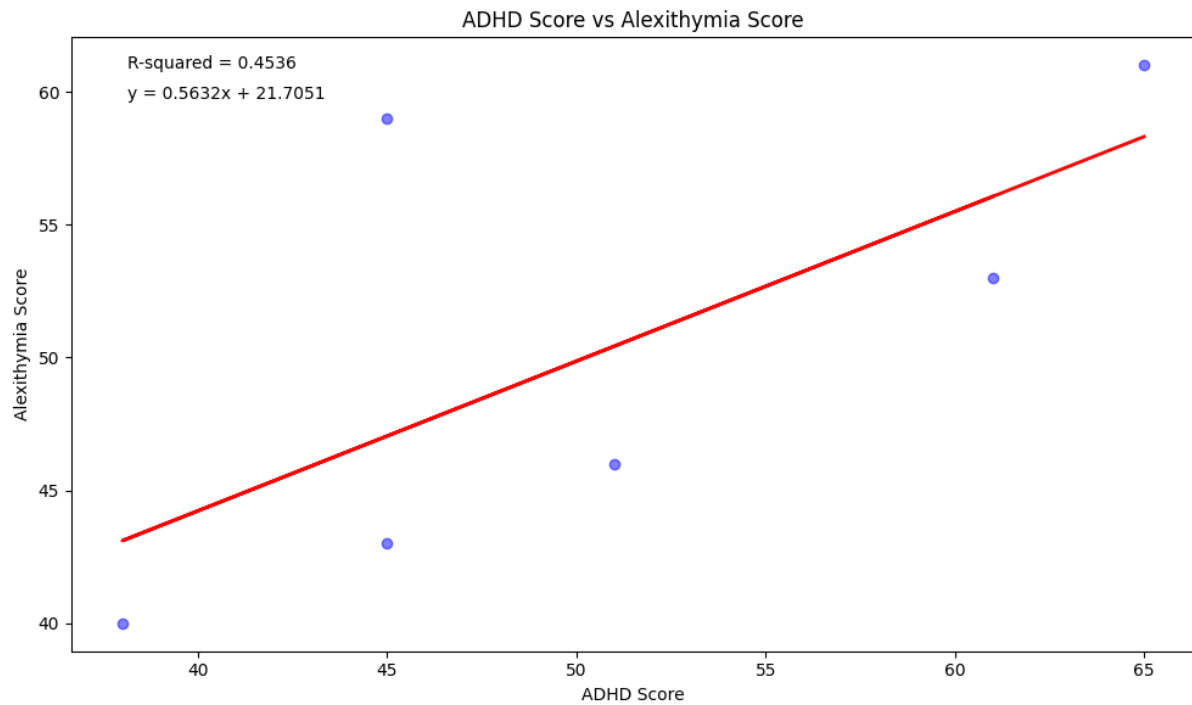


Figure 38 – Linear Regression of ADHD to Alexithymia

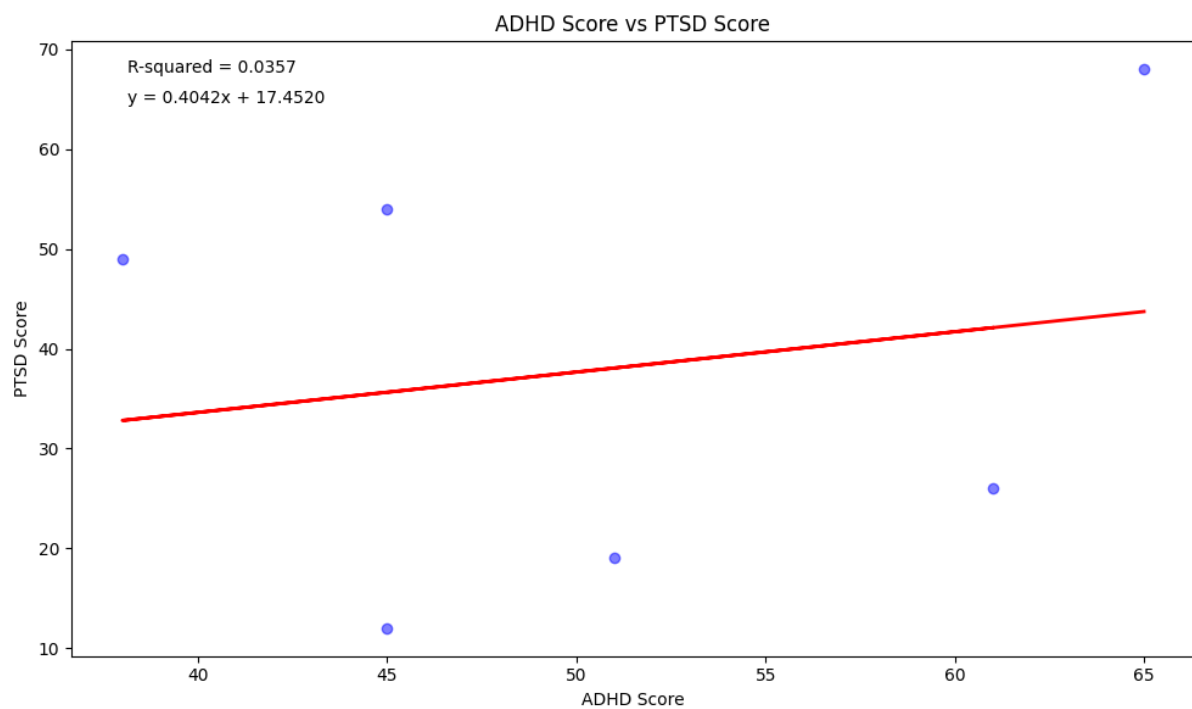


Figure 39 – Linear Regression of ADHD to PTSD

In figures 38, 39 we checked the link of ADHD to the other 2 conditions, which suggest there's some link between ADHD and Alexithymia, and a weak link to PTSD. Those graphs are weak evidence for links between the conditions since the sample size for ADHD scores was so little.

6. CONCLUSION AND FUTURE WORK

When predicting TAS-20 (Toronto Alexithymia Scale) questions using individual item predictions, error accumulation can become a significant issue. Each item's prediction error contributes to the overall error in the total score. Given that the TAS-20 comprises 20 items, the summation of these item-level errors can lead to a substantial deviation from the true total score. This error propagation occurs because any inaccuracies in predicting individual responses are compounded when these predictions are aggregated. Even small errors at the item level can lead to larger discrepancies in the total score, particularly if the errors are systematically biased in one direction.

This error accumulation is a significant factor contributing to the lowest accuracy observed in the method of predicting individual items. By focusing solely on predicting each item independently, we lose the broader context and "features" that the LLMs could utilize if they were assessing the score directly. The richness of information that comes from understanding patterns and relationships within the responses is diminished when the model is constrained to item-level predictions. This loss of contextual information leads to a less informed prediction process, thereby reducing overall accuracy.

Additionally, there is a paradox inherent in measuring alexithymia through the TAS-20 questionnaire. While the questionnaire is designed to quantify a complex psychological condition, its reliance on self-reported data can be problematic. Individuals with high levels of alexithymia may struggle to accurately self-assess and articulate their emotional experiences, leading to potential biases or inaccuracies in their responses. This reliance on self-reporting can undermine the very purpose of the questionnaire, as the condition it seeks to measure inherently affects one's ability to respond accurately. This paradox highlights the challenges and limitations of using questionnaires to measure complex psychological constructs, necessitating careful consideration of methodological approaches to ensure valid and reliable assessments.

In conclusion, our study found that the direct prompt method yielded the highest accuracy among all the approaches tested, with the lowest consistency error across predictions. This method's ability to maintain accuracy suggests it is the most reliable for predicting the TAS-20 scores. On the other hand, the sub-score prompt method achieved the highest confidence level by generalizing the item-level prompts and focusing the LLM on specific conditions related to the patient's mental state. This indicates that while direct prompts provide overall accuracy, sub-scores can offer deeper insights into particular psychological aspects. Future work could explore refining these methods further, potentially combining the strengths of both approaches to enhance prediction accuracy and reliability in assessing mental health conditions.

For instance, the direct prompt method provided a score of 3 with a confidence level of 8 for **spk_1**. The model identified several key indicators of alexithymia based on the language used. **spk_1** frequently relied on vague emotional terms such as "feeling bad" or "weird feeling," and showed difficulty in naming specific emotions or distinguishing between them. Additionally, **spk_1** focused more on describing physical sensations and external events rather than delving into their internal emotional state.

Despite some capacity for introspection, these patterns suggest a limited emotional range and a potential tendency toward alexithymia.

In contrast, another example of **spk_1** demonstrated emotional awareness by using detailed language to describe feelings of fear, anxiety, sadness, and stress. However, even in this case, the model detected a slight tendency towards alexithymia. This was evidenced by **spk_1**'s overreliance on physical sensations, such as a racing heart and trembling, to convey emotions and the use of general terms like "חרדה" (anxiety) and "לחץ" (stress) rather than more specific emotions. The model also noted that **spk_1** focused on external events rather than exploring the deeper emotional impact of those experiences. These examples illustrate how the direct prompt method can effectively assess the presence of Alexithymic tendencies by analyzing both the richness and specificity of emotional language used by the individual.

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8. APPENDIX A: QUESTIONNAIRES

A. TAS-20

Toronto Alexithymia Scale Questionnaire (TAS-20)

This questionnaire contains 20 questions. Answer each question by choosing a number from 1 to 5, according to the following scale:

1. Strongly Disagree
2. Disagree
3. Neither Agree nor Disagree
4. Agree
5. Strongly Agree

Answer the following questions:

1. I am often confused about what emotion I am feeling.
2. It is difficult for me to find the right words for my feelings.
3. I have physical sensations that even doctors don't understand.
4. I am able to describe my feelings easily.
5. I prefer to analyze problems rather than just describe them.
6. When I am upset, I don't know if I am sad, frightened, or angry.
7. I am often puzzled by sensations in my body.
8. I prefer to just let things happen rather than to understand why they turned out that way.
9. I have feelings that I can't quite identify.
10. Being in touch with emotions is essential.
11. I find it hard to describe how I feel about people.
12. People tell me to describe my feelings more.
13. I don't know what's going on inside me.
14. I often don't know why I am angry.
15. I prefer talking to people about their daily activities rather than their feelings.
16. I prefer to watch "light" entertainment shows rather than psychological dramas.
17. It is difficult for me to reveal my innermost feelings, even to close friends.
18. I can feel close to someone, even in moments of silence.
19. I find examination of my feelings useful in solving personal problems.
20. I look for hidden meanings in movies or plays.

The TAS-20 has 3 subscales:

Difficulty Describing Feelings subscale is used to measure difficulty describing emotions. 5 items – 2, 4, 11, 12, 17.

Difficulty Identifying Feeling subscale is used to measure difficulty identifying emotions. 7 items – 1, 3, 6, 7, 9, 13, 14.

Externally-Oriented Thinking subscale is used to measure the tendency of individuals to focus their attention externally. 8 items – 5, 8, 10, 15, 16, 18, 19, 20.

Scoring:

Range of scores: 20 to 100 (Higher scores reflect greater challenges or impairment.)

Each response is given a score on a Likert-type scale with 1 being strongly disagree and 5 being strongly agree.

Reverse scoring is used for items 4, 5, 10, 18, and 19.

The responses to all 20 questions add up to the overall alexithymia score, while the responses to each subscale factor add up to the score for that subscale.

B. IES-R

IMPACT OF EVENTS SCALE-Revised (IES-R)

INSTRUCTIONS:

Below is a list of difficulties people sometimes have after stressful life events.

Please read each item, and then indicate from the scale below, how distressing each difficulty has been for you during the past seven days with respect to the event.

Scale:

0. Not at All
1. A Little Bit
2. Moderately
3. Quite a Bit
4. Extremely

How much have you been distressed or bothered by these difficulties?

1. Any reminder brought back feelings about it
2. I had trouble staying asleep
3. Other things kept making me think about it.
4. I felt irritable and angry.
5. I avoided letting myself get upset when I thought about it or was reminded of it
6. I thought about it when I didn't mean to
7. I felt as if it hadn't happened or wasn't
8. I stayed away from reminders of it
9. Pictures about it popped into my mind.
10. I was jumpy and easily startled.
11. I tried not to think about it.
12. I was aware that I still had a lot of feelings about it, but I didn't deal with them.
13. My feelings about it were kind of numb.
14. I found myself acting or feeling like I was back at that time.
15. I had trouble falling asleep.
16. I had waves of strong feelings about it.
17. I tried to remove it from my memory.
18. I had trouble concentrating.
19. Reminders of it caused me to have physical reactions, such as sweating, trouble breathing, nausea, or a pounding heart.
20. I had dreams about it.
21. I felt watchful and on-guard.
22. I tried not to talk about it.

The IES-R has 3 subscales:

- Intrusion subscale is used to measure the occurrence of distressing thoughts, images, or memories of the traumatic event. 8 items – 1, 2, 3, 6, 9, 14, 16, 20.
- Avoidance subscale is used to measure efforts to avoid reminders of the trauma. 8 items – 5, 7, 8, 11, 12, 13, 17, 22.

- Hyperarousal subscale is used to measure symptoms of increased arousal and reactivity. 6 items – 4, 10, 15, 18, 19, 21.

Scoring:

- Range of total scores: 0 to 88. (Higher scores reflect greater challenges or impairment.)
- Each response is given a score on a Likert-type scale with 0 being Not at all, and 4 being Extremely.
- The responses to all 22 questions add up to the overall IES-R Score, while the responses to each subscale factor add up to the score for that subscale.

C. ADHD

Adult ADHD Self-Report Scale (ASRS-v1.1)

This questionnaire contains 18 questions. Answer each question by choosing a number from 1 to 5, according to the following scale:

1. Never
2. Rarely
3. Sometimes
4. Often
5. Very Often

Answer the following questions, Consider work/school, social and family settings:

1. How often do you have trouble wrapping up the final details of a project, once the challenging parts have been done?
2. How often do you have difficulty getting things in order when you have to do a task that requires organization?
3. How often do you have problems remembering appointments or obligations?
4. When you have a task that requires a lot of thought, how often do you avoid or delay getting started?
5. How often do you fidget or squirm with your hands or feet when you have to sit down for a long time?
6. How often do you feel overly active and compelled to do things, like you were driven by a motor?
7. How often do you make careless mistakes when you have to work on a boring or difficult project?
8. How often do you have difficulty keeping your attention when you are doing boring or repetitive work?
9. How often do you have difficulty concentrating on what people say to you, even when they are speaking to you directly?
10. How often do you misplace or have difficulty finding things at home or at work?
11. How often are you distracted by activity or noise around you?
12. How often do you leave your seat in meetings or other situations in which you are expected to remain seated?
13. How often do you feel restless or fidgety?
14. How often do you have difficulty unwinding and relaxing when you have time to yourself?

15. How often do you find yourself talking too much when you are in social situations?
16. When you're in a conversation, how often do you find yourself finishing the sentences of the people you are talking to, before they can finish them themselves?
17. How often do you have difficulty waiting your turn in situations when turn taking is required?
18. How often do you interrupt others when they are busy?

Scoring:

- Range of total scores: 18 to 90. (Higher scores reflect greater challenges or impairment.)
- Each response is given a score on a Likert-type scale with 1 being Never, and 5 being Very Often.
- The responses to all 18 questions add up to the overall ASRS Score.